

3
Book

Science TOPICS

FOR SENIOR STUDENTS

PETER GRIBBEN AND
WENDY ST GERMAIN



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CLASSMASTERS

Science TOPICS

FOR SENIOR STUDENTS

BOOK

3

**PETER GRIBBEN AND
WENDY ST GERMAIN**



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The three *Science for Senior Students* books present a wide variety of material aimed at students studying science subjects in Years 11 and 12. Many of the work sheets can be used as extension work for more able Year 9 and 10 students. The activities are designed to gain the interest of students and extend their knowledge, understanding and recall of science, something not always encouraged by conventional text books.

Over the three books in this series many branches of science are represented, including astronomy, biology, chemistry, ecology, geology and physics.

The broad range of subject areas is covered in a variety of styles to suit most ability levels. These include comprehension exercises, assignments, tests, graphing exercises, and calculations. The majority of topics are covered in a single page, with none being more than two pages.

The design highlights definitions, and key concepts, separate from the main text.

Many of the sheets can be used by teachers who have little prior experience in science.

In order to extend self-motivated students, and improve their fact-finding skills, most topics include a 'research' section. Outline answers to these are provided in the Answers sections.

The worksheets in *Science for Senior Students* are intended to:

- provide relevant material for class use to cover the absence of the regular teacher
- introduce new work in an easily assimilated manner
- provide homework or extension exercises
- review and revise specific topics prior to tests/examinations
- cover the 'gaps' left by standard text books.



When calculations or graphing work are included on a work sheet, these are indicated by the use of an icon/graphic.



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Water: Assignment

NAME _____ DUE DATE _____

Include a bibliography of the books and resources you use to complete this sheet.

- 1 Define 'arid' _____
- 2 Define 'brackish' _____
- 3 Define 'aquifer' _____
- 4 Define 'glacier' _____
- 5 Define 'hydrological' _____
- 6 Define 'permeable' _____
- 7 Define 'aqueous' _____
- 8 Name three substances present in sea water. _____

- 9 Why is each of the following added to water before it is supplied for drinking?
 - ozone _____
 - chlorine _____
 - fluoride _____
 - alum _____
- 10 Name one Australian area that is a 'wetland'. _____
- 11 Describe two physical characteristics associated with a wetland. _____
- 12 Name two organisms found in wetlands. _____
- 13 *Blue-green algae* is often mentioned in connection with Australian waterways. What is it and why is it a problem? _____



BIBLIOGRAPHY

Water Supplies

Water is the most common liquid on Earth and controls our lives. We drink it, wash in it, fish in it and grow crops with it. All living things need water. About two-thirds of your body is water. Every day you need two to three litres of water. This water can be taken in as food or drink. It replaces water lost in urine, sweat and when you breathe. People have been known to survive for more than a month without food, but only about a week without water.

Large amounts of water are also used in industry. Most of the water is used for cooling. A large power station uses about 5 megalitres of water per day. Because of this, power stations are often built near the coast or near rivers to ensure a large source of water for the cooling towers. Other major uses of water are for washing and to dissolve things. It takes about 10 litres of water to make 1 litre of beer, 200 litres to make 1 newspaper, and 50 000 litres to make a medium-sized car. Per day, a person might use 70 litres flushing the toilet, 50 litres for personal washing, 25 litres for clothes washing, 15 litres for dishwashing, 12 litres for gardening and carwashing, 5 litres for cooking and 3 litres for drinking.

Before water reaches us, it has been purified by the water authorities but, originally, it will come from either surface water (rivers and lakes) or ground water in underground wells. If we trace the water supply back even further, then the source of all this water is rain.

Sewage is the main cause of water pollution. If the amount of sewage is small, then bacteria in the water can break it down into less harmful materials such as carbon dioxide, nitrates and water. If the amount of sewage is too large, then the bacteria use up all of the oxygen dissolved in it as they feed on the sewage. Once the oxygen concentration gets too low, most of the living organisms in the water (including the bacteria) die and the water becomes cloudy and smelly. In order to avoid this pollution, sewage is treated before it is returned to rivers and the sea. The sewage is pumped into large tanks and mixed with air (aerated) so that it can be more easily decomposed by bacteria.

The heat from power stations can be a water pollutant. The water in cooling outlets can be 10°C higher than normal. This higher temperature reduces the amount of oxygen dissolved in the water. The reduced oxygen content can be harmful to some fish such as salmon or trout, although other species thrive on the warmer water. So industrial uses for water can indirectly affect the balance of species in an ecosystem.

Other sources of pollution include fertilisers, industrial chemicals, pesticides, oil and detergents.

- 1 Draw up a table showing the daily use of water by a person.
- 2 How much water might a large power station use per day?
- 3 What is the main source of water pollution?
- 4 Can hot water dissolve more or less oxygen than cold water?
- 5 How much water is needed to make 375 mL of beer. Show your working.
- 6 How is sewage treated before it is released into the environment?
- 7 A car plant produces 50 000 vehicles a year. How much water would the factory use per day? Show your working.
- 8 Why might a 'fish farm' be set up near the cooling outlets of a nuclear power station?
- 9 What happens to water when the level of dissolved oxygen falls to a low level?
- 10 In some parts of Africa, each person must survive on less than 10 litres per day. How does that compare with our average daily use?



RESEARCH

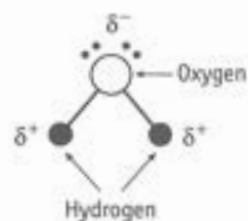
- 11 Find out what the cost of water is for domestic customers in your area.
- 12 Untreated sewage can cause health hazards and lead to diseases such as cholera. Find out the effects of, and medical treatment for, cholera.

Water: Comprehension

Water is a liquid at room temperature. It can also exist as a solid in the form of ice, and as a gas in the form of steam. Ice melts (and water freezes) at 0°C . Steam condenses (turns to liquid water) and water boils at 100°C . Because water is a liquid at room temperature, it has no shape of its own. It takes up the shape of its container. Pure water has no colour, smell or taste. One millilitre (mL) of water has a mass of 1 gram (g). Its density is one gram per millilitre (1 g mL^{-1}). Density is the amount of mass that is in a certain volume, or $\text{mass} \div \text{volume}$.

The reason that popcorn 'pops', is that there is water inside the maize kernel (corn). When you heat it, the water turns to steam and expands. This causes the pressure to increase and the corn to explode.

Scientists sometimes refer to water as ' H_2O '. It is a 'v-shaped' molecule. The oxygen atom is slightly negatively charged and the hydrogen atoms are slightly positively charged. This difference in charge helps water dissolve so many substances.



- 1 There are many liquids other than water. Name three liquids that don't contain water.

Other liquids such as tea, milk and fruit juice all contain water. In fact they are mostly water with small amounts of other things added. These give the water colour and flavour.

If you put some water in a test tube, then add a small amount of sugar (sucrose) and shake it, the sugar seems to disappear. It is said to be *soluble* in water, it has *dissolved* and formed a *solution*.

- 2 What is the *solvent*?
- 3 What is the *solute*?

The surface of the Earth is more water than land. About two-thirds (67%) of the Earth's surface is covered with water. Most of this water is in the form of seas and oceans. These are different from inland lakes and rivers. Seas and oceans are salty (saline), meaning they contain dissolved salt (sodium chloride, NaCl). Water is present in the biosphere, lithosphere, hydrosphere and atmosphere. Moisture (humidity) is measured using a hygrometer, usually a 'wet and dry bulb' thermometer. Humidity is expressed as a percentage. When humidity is great, rain falls. Rainfall is measured using a rain gauge. The units used are millimetres (mm) or centimetres (cm).

The *biosphere* is the part of the Earth that can support life, extending from the ocean depths to about 10 km into the atmosphere. The *lithosphere* is the Earth's crust. The *hydrosphere* is all water that is found in, on or enveloping the Earth, including water vapour in the atmosphere. The *atmosphere* is the air that surrounds the Earth. Weather is determined by the environment. Climate is a pattern of weather. Although humans use a lot of water, it doesn't run out. The reason for this is the *water cycle*. The sun provides energy for this process.

Clouds are part of the water cycle. There are three main types. *Cirrus* clouds are wispy, high-level clouds. *Cumulus* clouds are large, fluffy clouds found lower in the atmosphere. *Stratus* clouds are horizontal layers of cloud. If the word *nimbus* is included in a cloud's name, it means that the cloud looks dark and is likely to give rain.

Nearly two-thirds (67%) of the Earth's surface is covered in water. It rains from the sky and pours into the seas and rivers.

Water: Comprehension

The Sun warms up water (for example, the sea) that evaporates and goes into the air as water vapour. Water vapour also rises into the air from water evaporating from trees (this is called *transpiration*), breathing (*respiration*) and burning (*combustion*). The water vapour then condenses to make clouds. These clouds rise when they get to mountains, then cool down and the water falls as rain. This rain flows down mountains and hills and joins together to form rivers. The rivers lead to the sea and the process starts all over again.

As well as all the water on the Earth's surface, living things contain a lot of water. The body of a 15-year-old contains about 35 litres of water. Each day the human body loses about 2.5 litres of water. Most of this is lost as urine and sweat. The rest is breathed out as water vapour. This loss must be replaced by food and drink. Without water, people can die in as little as 4 or 5 days. Apart from drinking, we use water for washing, cooking and flushing toilets. A person uses, on average, 140 litres in a day.

Once water has been used, it becomes sewage. It has grease, dirt, detergents, unused food and toilet waste in it. This sewage must be treated before we can be rid of it. Many factories need large amounts of water and are often built close to rivers. They take water from the river. When they have finished with the water, they discharge the waste back into the river. By law, they must try to clean up their waste before they discharge it into the river.

- 4 Using the information contained in these sheets, draw a labelled diagram of the *water cycle*.
- 5 State the name and formula of the chemical that forms the majority of solids dissolved in seas and oceans.
- 6 How much water is in the body of the average 15-year-old?
- 7 How much water do we each lose per day from our body?
- 8 How much water does the average person use per day?
- 9 What is the *biosphere*?
- 10 What is the difference between *fresh* and *saline* water?
- 11 List the names and symbols of the elements contained in water.
- 12 What percentage of the Earth's surface is covered in water?
- 13 Water has a density of 1 g mL^{-1} . What does this mean?
- 14 What is 'sewage'?

In ancient times philosophers (thinkers or scholars) said that water (along with earth, air and fire) was a basic element. This belief continued until about 250 years ago.

- 15 Is water an element? Give a reason for your answer.
- 16 Draw a labeled diagram of a water molecule.
- 17 Name three types of cloud.



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- 18 Explain, with the aid of a diagram, how seas and oceans have become 'salty'.
- 19 Some farmers still refer to rainfall in *points*. These are non-metric units. How does this measurement compare with current (metric) units?
- 20 What is 'brackish' water?
- 21 Explain why high humidity makes people feel uncomfortable.



Water and Life: Assignment

NAME _____ DUE DATE _____

All living things contain water (usually being somewhere between 66% and 95% water). A typical male human adult's body will be about two-thirds water.

- 1 Assuming that tomatoes are 94% water, calculate how much water there is in 1 kg of tomatoes. Show your working. _____

Water transports many substances around the body. This is because it is such a good solvent. Water represents a major constituent (over 80%) of blood. Blood carries nutrients and waste products around the body. It carries oxygen from the lungs to cells and waste products, such as carbon dioxide, to the lungs. When food is digested it is broken down into simpler, soluble products and these are carried by the blood to cells to provide energy and raw materials.

- 2 Give the chemical formula for water. _____
- 3 Give the chemical symbol for oxygen. _____
- 4 Give the chemical formula for carbon dioxide. _____

Water is a chemical reactant in photosynthesis. In this process, green plants use sunlight to cause carbon dioxide to react with water to form glucose ($C_6H_{12}O_6$) and oxygen.

Respiration occurs in animal cells, producing heat (and waste products). Water has a high specific heat, which means it can store a large amount of heat. Blood can carry the heat from respiration around the body. Water also helps the body to keep cool. When a person sweats, water evaporates from the skin, taking heat from the body in the process.

Large bodies of water tend to have steadier temperatures than the land. Coastal regions have milder climates than inland areas. This is because of water's great capacity to store heat. When layers of ice form on the surface of water in very cold climates, the ice acts as an insulator for the water beneath.

Water provides a habitat for many organisms. It also contains dissolved oxygen for aquatic plants and fish to breathe. Water also provides support (by displacement) for large mammals such as whales, who have great problems with their body mass when out of the water.

- 5 Define 'habitat'. _____
- 6 What is meant by 'aquatic'? _____

Water can affect our environment — the amount of water present will affect what lives in a particular area. Some animals drink very little, getting all or most of their water requirements from plants, for example, koalas gain water from gum leaves. Other organisms need a lot of water, for example, plants such as water lilies need to grow in water.

Water can cause weathering (breaking down) of rocks. If it rains, water can collect in the cracks in rocks. If the water then freezes, it expands and can crack the rocks. Once rock has been broken down into small pieces it can be easily removed, so erosion occurs. Running water (in rivers) or waves can also alter the shape, colour or texture of rocks. Glaciers of ice, although they move slowly, can move millions of tonnes of earth and rocks.

Water and Life: Assignment

Water's great heat capacity is used to provide cooling for car engines, coal-fired power stations, and some home heating and cooling systems.

Water is a natural resource used by humans and other organisms. As well as for industry and agriculture, water is used for recreational activities.

- 7 Name three recreational uses of water. _____

Because of its physical properties, water plays an important part in the survival of all organisms. Some important characteristics include its:

- role as a solvent for so many substances
- ability to store and move heat
- expansion on freezing
- existence as a solid, liquid or gas under normal conditions.

- 8 A male teacher weighed 78 kg. What is the amount of water in his body? Show your working.

- 9 Write a balanced equation to describe photosynthesis.

- 10 Which physical property of blood allows it to transport food around the body? _____

- 11 Which physical property of blood allows it to transport heat around the body? _____

- 12 Give two ways in which water can wear away rocks. _____

- 13 What does 'constituent' mean? _____

- 14 Why can ice forming on the surface of a lake in winter help fish in the lake survive?

- 15 Why does perspiration (sweating) help keep us cool? _____

- 16 Why do water habitats have temperature extremes less than those in terrestrial habitats?

Say whether each of the following statements is true or false.

- 17 About 20% of blood is water. _____
- 18 Gum leaves in the diet of koalas gives them the water they need. _____
- 19 Weathering is the movement of broken-down pieces of rock. _____



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- 20 Find out how the water requirements of kangaroos compare with those of cattle or sheep. Comment on your figures.
- 21 Using whales as an example, explain what is meant by 'displacement of water'.
- 22 Give an example of a plant, not mentioned on the sheet, which lives in an aquatic environment.



Water:

Some facts and figures

Aldonga is a typical inland Australian town. Its residents use, on average, 350 kilolitres of water per household per year. Aldonga is allocated 9500 megalitres (million litres) from the local water authority. In one year Aldonga used 9375 megalitres of this allocation.

- 1 What percentage of the water allocation was used? Show your working.
- 2 Using the figures above, how many households are in Aldonga? Show your working.

In order to encourage local residents to be more careful with their water, the local council has a two-tier rating system. The first 250 kilolitres used by a household are charged at 15 cents per kilolitre and subsequent kilolitres are charged at 50 cents a kilolitre.

- 3 Calculate (showing working) the water bill for a family using the average amount.
- 4 Calculate the bill for 450 kilolitres per year.

As part of a campaign to raise awareness of water use and quality of local waterways, all households in Aldonga were delivered a pamphlet which mentioned the following hints.

- Turn off the tap when you brush your teeth.
- Wash your car with a bucket and a sponge.
- Wash your car on a grass surface.
- Only use dishwashers and washing machines with a full load.
- Use low phosphorus detergents.
- Use a timer on garden sprinklers.
- Replace washers on leaking taps.
- Collect and bury pet droppings.
- Don't pour chemicals and paints down the drain.
- Use mulch around fruit trees.



- 5 List the hints above that are ways to save water.
- 6 List the hints above that are ways to keep local waterways pollution-free.

A couple were looking at washing machines and one consideration was water use. They found that most top loaders used about 150 litres of water for a 5 kg load of washing, whereas front loaders used about 105 litres for the same quantity. They estimated they had three full loads of washing per week.

- 7 Calculate how much water the couple would save in a year using a front loader washing machine compared to a top loader.
- 8 Name two other things they would take into consideration apart from water usage.

An Aldonga school student decided to study the water content of a potato as part of her school science project on water. Her plan was to dry a potato and work out how much water was lost. She peeled and weighed the potato. She then placed the potato in a warm dry place and weighed the potato at regular intervals until the weight remained constant. Her results are shown below.

Day	1	2	5	7	10	12	15	17	19	21	24	26	28	30
Mass (g)	95	87	66	52	41	38	36	34	32	31	30	29	28	28

- 9 Draw a line graph using the figures in the table. Plot time on the x-axis and mass on the y-axis.
- 10 What mass of water was present in the potato?
- 11 Give two assumptions you made in giving your answer to Question 10.
- 12 What percentage of a potato is water?
- 13 Give two ways in which the student could improve the accuracy of the experiment.

Properties of Water

When a substance dissolves in water to form a solution, the substance is the *solute* and the water is the *solvent*. Water is a very important solvent — it is sometimes called the 'universal solvent'.

- 1 Seawater is a solution of (mainly) sodium chloride. Name the solute and solvent in seawater.

Water is present in the biosphere, the lithosphere, the hydrosphere and the atmosphere. Water is essential for life to exist. Cells are the basic units of life. Cells contain water which acts as a solvent and a raw material in metabolism. Human bodies contain between 55% and 70% of water.

A habitat is where an organism lives and can be aquatic or terrestrial. Aquatic habitats can be fresh or saline (salty) and have more constant temperatures than terrestrial habitats. The smaller variations in temperature may make aquatic habitats more favourable than land for life to survive.

- 2 Explain what is meant by 'metabolism'.
- 3 Explain what is meant by 'aquatic'.
- 4 Explain what is meant by 'terrestrial'.
- 5 Explain what is meant by 'saline'.

Water is an weathering agent of rocks, both as a liquid and a solid.

- 6 Explain how water can break down rocks.

Water is a natural resource for humans and other organisms. We use water for recreation, irrigation, household use and industrial use.

- 7 Name two ways that water is used by households.
- 8 Name two ways that water is used by industry.

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}}$$

Density (specific gravity) is the ratio of mass of an object compared to the space it occupies. The density of water is sometimes quoted as 1.00 g mL^{-1} .

- 9 What is this density expressed in kg m^{-3} (kilograms per cubic metre)? Show your working.

Water behaves in a strange way when cooled; it has its maximum density at 4°C . It also expands upon freezing. Ice at 0°C has a density of about 0.92 g mL^{-1} , which is less than the density of water and is why ice floats on water.

The way that water molecules line up at different temperatures is the reason for the different densities at these temperatures.

Pure water at STP (Standard Temperature and Pressure)/SLC (Standard Laboratory Conditions) freezes at 0°C and boils at 100°C . When dissolved substances are present, these 'fixed points' change. Motorists add a substance called ethylene glycol (ethan, 1-2 diol) to the water in their car radiators. Manufacturers label this substance as 'Antifreeze-Antiboil'. It raises the boiling point and lowers the freezing point of the water. This means that the fluid in the radiator is less likely to boil in warm weather or to freeze in cold weather.

- 10 What might happen to a car radiator containing only water if the temperature dropped to, for example, -5°C ? Explain your answer.
- 11 Why do icebergs float?



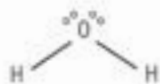
RESEARCH

- 12 What conditions are described by STP and SLC?
- 13 An important measure of water quality is Biological Oxygen Demand (BOD). Explain what this term means. (Note: sometimes referred to as Biochemical Oxygen Demand.)
- 14 Explain why additives can raise the boiling point of water.

Forces Between Water Molecules

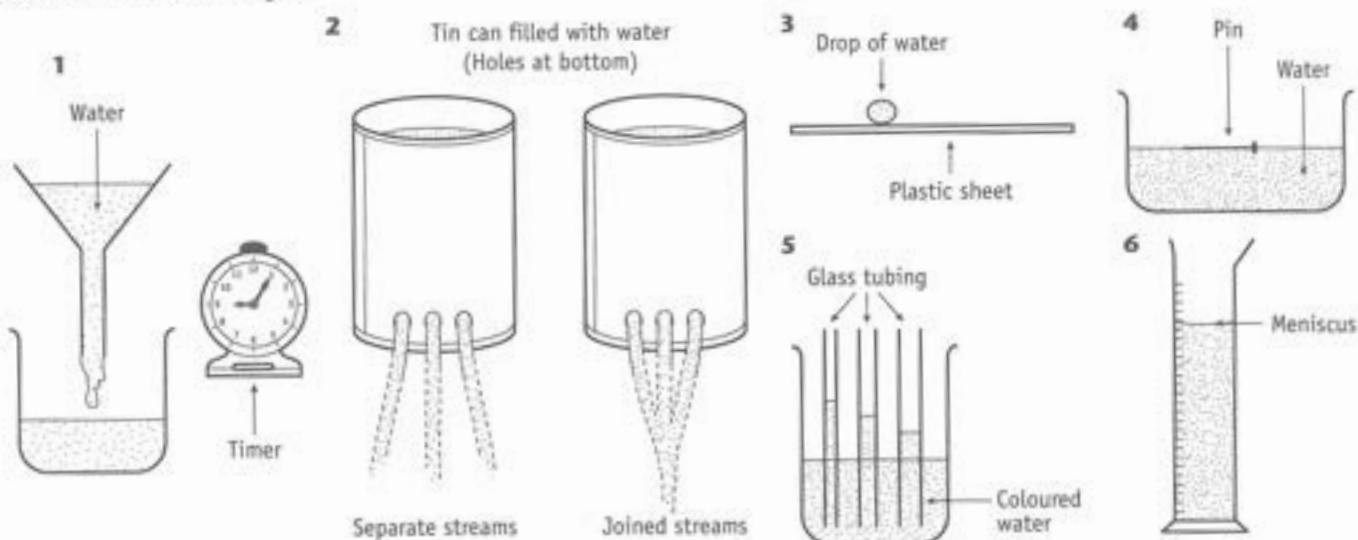
The attractive forces between molecules (intermolecular forces) affect water's properties, for example:

- surface tension
- adhesion
- cohesion
- viscosity.



- **Surface tension** is caused by intermolecular forces pulling on the individual water molecules on the surface of the liquid. It results in an elastic 'skin'.
- **Adhesion** is the attraction between two different surfaces in contact with each other, for instance water and a glass container. It causes *capillary action*, which cause a liquid to be drawn up a narrow tube.
- **Cohesion** is the force of attraction between liquid molecules within the same body. It causes liquid surfaces to become as small as possible.
- **Viscosity** is a measure of how a liquid resists flowing.

The diagrams below (not to scale) show a variety of activities demonstrating various properties of water. State which property of water (surface tension, adhesion, capillary action, cohesion or viscosity) is being shown in each example.



- 1 Timing how long water takes to flow. _____
- 2 When water pours out of a can, the streams can be joined into one. _____
- 3 Water forms droplets on a non-absorbent surface. _____
- 4 Floating a steel pin on water. _____
- 5 Water moving to different levels up tubes with different diameters. _____
- 6 A curved surface of liquid in a tube. _____
- 7 What is meant by 'elastic'? _____



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- 8 Water is a fluid. Define 'fluid'. _____
- 9 Explain why water expands when it freezes. _____

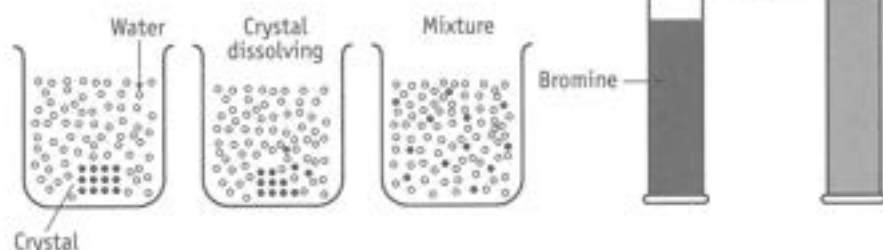
Diffusion and Osmosis

Particles in liquids and gases can move. In order for organisms to survive, substances must move in and out of cells. Most materials travel around the body in solutions.

- 1 What is the solvent for the solutions in the human body?

The diagrams below show simple examples of diffusion. The three beakers represent what happens when a crystal dissolves in water. The two measuring cylinders show the movement of bromine vapour (a heavier-than-air gas).

- 2 Use the particle theory of matter to explain what is happening in the diagrams.
- 3 What is the formula for a bromine molecule?
- 4 What does 'kinetic' mean?

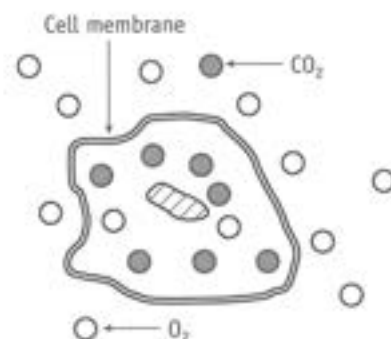


Diffusion is the movement of particles from an area of higher concentration to an area of lower concentration. It stops when particles are evenly distributed.

Diffusion allows the movement of gases, such as oxygen and carbon dioxide, in and out of cells through cell membranes. All cell membranes are semi-permeable, so they allow some substances to pass through but not others.

The diagram on the right represents a cell surrounded in a fluid. (To simplify things, water molecules are not shown).

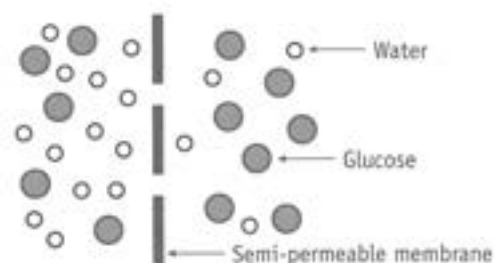
- 5 Copy the diagram.
- 6 On your diagram, indicate how the oxygen and carbon dioxide molecules will move. Assume they are small enough to pass through the membrane.
- 7 When will movement of particles be at a minimum?



Osmosis is the movement of solvent molecules through a semi-permeable membrane so as to reduce the concentration of the solute on the other side.

Osmosis is a type of one-way diffusion. It involves movement of solvent molecules only. The diagram below shows molecules of glucose ($C_6H_{12}O_6$) dissolved in water. The concentrations are different on either side of the membrane.

- 8 Copy the diagram.
- 9 Which side has the higher concentration of glucose?
- 10 On your diagram, indicate how particles will move to even out the concentrations.
- 11 What is the characteristic of the membrane that allows some particles to pass through and not others?
- 12 Give the names of the elements present in a glucose molecule.
- 13 Give two similarities and two differences between *diffusion* and *osmosis*.





Substances Dissolving in Water

Many of the chemical reactions that we carry out occur in solutions. There are two parts to a solution, a *solute* (which is the substance being dissolved) and a *solvent* (which is the substance in which the solute is dissolved). The solvent we use most is water.

Concentrations of solutions can be measured in several ways, for example:

- mass/mass
- mass/volume.

Mass/mass is a measure of how much of a solute is dissolved in a given mass of solution. A solution of magnesium chloride (MgCl_2) could be described as 5 g/100 g. This would mean that 5 g of the solute (magnesium chloride) had been dissolved in 100 g of solution.

Mass/volume is a measure of how much of a solute is dissolved in a given volume of a solution. To make comparison easier, the volume used is usually a litre. A solution of magnesium chloride stated to be 50 g L^{-1} would have 50 grams of solute dissolved per litre of solution.

- 1 What is a solution of 50 g in 100 mL expressed in grams per litre (g.L^{-1})?
- 2 What is a solution of 5 g in 500 mL expressed as g.L^{-1} ?
- 3 What is a solution of 25 g in 250 mL expressed as g.L^{-1} ?
- 4 What is a solution of 75 g in 2000 mL expressed as g.L^{-1} ?

Solubilities of solids are usually quoted at a specific temperature. The solubility of most solids *increases* with a rise in temperature.

- 5 Give two other ways in which a solid can be made to dissolve more quickly.
- 6 Explain what is meant by the term, 'foreign matter'.

Pollution affects the quality of water and can make it unfit for human use. Especially worrying are the 'heavy metals', such as cadmium, mercury and lead.

- 7 Write the symbols for the elements mentioned as 'heavy metals'.

The heavy metals and solutions of their compounds are highly toxic — their effects are often cumulative. Mercury is used as a catalyst in some chemical reactions. If, for instance, mercury and its compounds make their way into waterways, these substances can be taken up by fish and other aquatic organisms. The living creatures do not die immediately but accumulate mercury compounds in their bodies. Other living creatures eat them and the concentration of mercury compounds rises in each level of the 'food chain'. Humans who have eaten fish (and other food) with high mercury levels have been affected. Major symptoms of mercury poisoning are brain, kidney and liver damage. Mercury and its compounds are not biodegradable.

- 8 What is meant by 'toxic'?
- 9 What is meant by 'cumulative'?
- 10 What is meant by 'food chain'?
- 11 What is meant by 'biodegradable'?
- 12 What is meant by 'catalyst'?



Pollution of water is caused by contamination by foreign matter such as microorganisms, chemicals, industrial waste, farming wastes, household waste and sewage that reduce the quality of the water.



RESEARCH

- 13 Find out how the solubility of gases changes with temperature when water is the solvent.



Koalas and Other Australians

The koala (*Phascolarctos cinereus*) is only found in Australia. It is an arboreal marsupial (not placental) mammal. It feeds almost exclusively on the leaves and flowers of certain (about 20) species of eucalypts. A koala will seldom drink. It obtains most of its water from the leaves it eats. It is lethargic and often stays in the same tree for days at a time. The koala grows to about 75 cm in length and has a life span of approximately 12 years in the wild, but can live for up to 20 years in captivity.

- 1 Explain what is meant by 'arboreal'.
- 2 Explain what is meant by 'lethargic'.

An area is being studied with the possibility of it being used as a reserve for koalas. The koalas in this region only eat the 'grey gum' species of eucalypt. A survey was made of the area counting the number of suitable trees in 1 km² areas chosen at random. The results are shown in the table.

Area	Number of grey gums
A	4
B	1
C	3
D	5
E	2

- 3 Calculate the average density of grey gums in the reserve.
- 4 Assuming that each koala needs three gum trees for food and shelter, what would be the minimum area needed for the reserve to support a population of 100 koalas?
- 5 Read the following passages and give definitions for the terms written in **bold**.

The Tasmanian tiger or Tasmanian wolf (*Thylacine*) was a carnivorous **marsupial** that preyed on kangaroos. The major reason for its **extinction** in mainland Australia was likely to have been competition from the dingo. The dingo (*Canis dingo*) is similar to domestic dogs. It is thought to have **originated** in Asia and to have accompanied humans coming to Australia over 30 000 years ago. While dingos are primarily carnivores they eat almost anything, including **carrion**. After Europeans arrived in Australia, dingos started to prey on the **introduced** sheep and rabbits rather than on native wallabies and kangaroos. Dingos rarely hunt in large groups, it is more usual for them to **forage** by themselves, in pairs, or sometimes as a small family pack.

Tiger quolls, sometimes called 'native cats', are **carnivorous** marsupials. They are found in New South Wales, Tasmania, Queensland, the south-east of South Australia and the east of Victoria. They are becoming rare on the mainland and some scientists think quolls are in danger of dying out.

Tiger quolls have a rusty brown coat with white spots on the body and tail. Their fur underneath is cream-coloured. Male tiger quolls grow up to 1.3 metres long and can weigh up to 7 kilograms. Females are smaller.

The tiger quoll's natural **habitat** is forest and thickly wooded areas, where they live on bird's eggs, small mammals, reptiles, invertebrates and dead animals. Following a **gestation** period of 21 days, female tiger quolls give birth to between one and six young. After about ten weeks the **offspring** become too large to fit in the mother's pouch and she leaves them in the nest or takes them with her, clinging to her back. Quolls become independent when they are about four months old.

Quolls are generally **nocturnal**. In the wild they live from three to five years, but can live up to six years in captivity.



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- 6 Explain the major differences between placental and marsupial mammals.
- 7 Give two facts about quolls not already mentioned.
- 8 Give a brief description of the appearance of the thylacine.
- 9 Why might animals live longer in zoos than they do in the wild?

Kangaroos

Kangaroos are furry, marsupial mammals only found in mainland Australia and bordering islands. There are over fifty species of kangaroos. They hop on their hind (rear) legs. Their front legs are short and the five digits ('fingers') are of different sizes. Their long, powerful hind legs have tendons that act as springs for powerful leaping. A large kangaroo can jump over six metres in one bound and reach a top speed of nearly 50 kilometres per hour. Kangaroos can jump nearly two metres high.



Kangaroos use their long, muscular tails for balance when hopping and support when walking or sitting. They are not normally aggressive, but can cause harm when they feel threatened, hitting out with their forepaws and kicking with their muscular rear legs. Apart from humans and dingos, kangaroos have no natural enemies. They were once hunted extensively for their meat and coats. Kangaroo hunting was once a popular sport in Australia, with hundreds of thousands killed each year. Some are still hunted in areas where they are abundant or are a nuisance.

Kangaroos are marsupials and females have abdominal pouches. A newly born kangaroo (2.5 cm long) makes its way slowly into the pouch. It does this by tracing a path of fur licked by the mother kangaroo. The 'joey' remains in the pouch until it is six to nine months old. This pouch usually contains four mammary glands — only two of these at a time operate to give milk. When the joey permanently leaves the pouch, it still stays with its mother and continues to obtain milk by putting its head into her pouch, usually for another six to nine months. Kangaroos can live for up to seven or eight years.

Kangaroos are also described as *macropods*, meaning they have large feet. The best known and largest kangaroos are the giant, or great grey, kangaroo (*Macropus giganteus*) and the red kangaroo (*Macropus rufus*). Red kangaroos tend to be found in the deserts and dry grassland of central Australia. Grey kangaroos are mainly inhabitants of the grasslands and forests of eastern and southern Australia. The male adults of both species can stand over one-and-a-half metres in height, not counting the tail, which is over a metre long. Females tend to be smaller. Large kangaroos are herbivores. Graziers have claimed that kangaroos damage their paddocks, but some surveys have shown that sheep and cattle eat different foods.

Smaller kangaroos, called wallabies, are usually brighter in colour than the larger ones. Many of these species are rabbit-sized. The red-necked wallaby inhabits bushes, whereas the rock wallabies live in cracks in rocks. Unlike most other kangaroos, the rock wallabies are nocturnal (come out at night). Most kangaroos are terrestrial — the only arboreal (living in trees) kangaroos are small tree kangaroos which have front legs almost as long as their hind legs.

- 1 How many types of kangaroo are there?
- 2 *Macropus giganteus* is the scientific name for what type of kangaroo?
- 3 How far can a large kangaroo jump?
- 4 Give two uses that a kangaroo makes of its tail.
- 5 How big is a full grown male red kangaroo, from head to tail?
- 6 What do kangaroos eat?
- 7 Where do most grey kangaroos live?
- 8 What is a mammary gland?
- 9 Give two differences between kangaroos and wallabies.
- 10 Define 'nocturnal'.
- 11 Define 'terrestrial'.
- 12 Define 'arboreal'.
- 13 Define 'abundant'.



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- 14 Write a report on a species of marsupial not already mentioned.

Noxious Weeds

Noxious weeds are seen as a growing problem in most States. They affect not only farmers and graziers, but impact on urban environments as well.

- **Noxious:** *physically harmful to organisms.*
- **Weed:** *a plant that is growing where it is not wanted.*

'Paterson's Curse' (*Echium plantagineum*) is a well-known plant. It is an introduced species which came from the Mediterranean area in the 1850s and is common throughout southern parts of Australia. Plants can produce in the region of five thousand seeds each. These seeds can remain dormant (inactive) for up to five years or can germinate at any time of the year if conditions are suitable.

1 In general, what conditions are needed for a seed to germinate?

Paterson's Curse seeds have rough seed coats and can be spread by:

- contaminated seed used for sowing crops
- run-off water
- being carried on coats of animals, such as sheep and cattle
- being eaten by stock and passed out in droppings.

Paterson's Curse is designated as a weed because it:

- replaces native plants
- chokes out pasture plants
- reduces soil fertility
- is poisonous to grazing stock, especially horses and pigs.

'Salvation Jane'
or
Paterson's Curse



Estimates vary, but it is possible that Paterson's Curse costs Australian agriculture over \$100 million per year.

There are selective herbicides available to deal with Paterson's Curse, but these can adversely affect legumes.

2 What is meant by a 'selective herbicide'?

A technique called 'Spray-graze' can be used to reduce levels of herbicide. The herbicides are applied before the plant has reached maturity and, after a withholding period, the pasture is stocked with several times the normal level of animals.

3 What is a 'withholding period'?

This heavy grazing will reduce Paterson's Curse to a manageable level, but care must also be taken that stock do not graze for extended periods of time on the treated pasture. Sheep and goats are least affected by the plant.

4 Name two animals which are severely affected by Paterson's Curse.

One increasingly popular way of controlling Paterson's Curse (and other weeds) is by using its natural enemies. Although there are many insects from Europe which eat Paterson's Curse, it has taken many years for approval to be gained or any of these to be released in Australia.

5 Give two reasons why people have been reluctant to introduce new species of insects.

Biological control never completely eradicates the targeted weed but it does reduce it to manageable levels.

6 Imagine you are a scientist working for Australian quarantine authorities. Another Government research department comes to you with a species of overseas insect which they want to import to control a weed that is causing problems. Outline the steps you could follow before giving approval for general release of this organism.

7 Name three ways by which seeds of Paterson's Curse can be spread.

8 Why might some States and Territories declare Paterson's Curse to be a noxious weed?

Concern Over Crayfish

Read the following magazine article and answer the questions.

Crayfish are a prized catch and command high prices both in Australian and international markets. Local fishermen are showing concern about recent dramatic drops in the number and size of their catch of crayfish. They are concerned with the effect that it will have on their livelihood if the number of crayfish currently being caught remains at a low level.

A representative of the local commercial crayfishermen's association raised concerns about the state of their industry.

'Many of our members are doing very badly. Our catch is a fraction of what it was a year ago. We still have the same overheads. We're not even covering our costs at the moment and many of the operators will go out of business unless something is done. We're not marine scientists but between us we've got a lot of knowledge and we reckon we know what's wrong. We've had a lot of problems with octopuses lately. They are raiding our pots and taking the crayfish. Some of the others in the industry have been catching heaps more octopuses than ever.

Sharks keep the numbers of octopus to a reasonable level, I've seen nurse sharks contain up to a dozen octopuses when they've been cut up. Over the last few years they've taken too many sharks. The wholesale netting that some use isn't discriminating enough, it caught all sorts of sharks. The non-commercial species have been pulled out as well as the stuff that can be sold in the market. They've even used shark as bait.'

A spokesperson for the Marine Fisheries Department supplied the following information.

'Crayfish are arthropods. They have an exoskeleton, jointed limbs and resemble lobsters. There are strong claws on the first of their five pairs of walking legs. They are scavengers and feed upon decaying (or sometimes live) animal or vegetable matter. They play an essential part in marine ecosystems by returning organic substances to the various cycles. Crayfish have been commercially fished in this area for a considerable period of time and there is a steady demand for these

and other good quality crustaceans. We have no information to hand that catch numbers and quality have diminished in recent months.

Octopuses are cephalopods (head-footed), that is the mouth is surrounded by limbs. They are carnivorous molluscs which feed on other molluscs and crustaceans. Some species are considered edible by humans. We are not aware that numbers have significantly increased.

Sharks are cartilaginous fish and can be aggressive carnivores or docile feeders on plankton, depending on species. Sharks are also scavengers and keep down waste and remove sick or injured marine organisms from the general population, which maintains the fitness of these species. Some parts of the fishing industry see sharks as competitors, others catch shark to sell as an alternative to more traditional fish varieties.'

- 1 Draw up a food web using the information contained in the article.
- 2 Which organism does the fisherman blame for the drop in numbers of crayfish caught?
- 3 What might the fisherman suggest as a solution to the problem?
- 4 Give another possible reason to explain why less crayfish are being caught.
- 5 Imagine that you are a scientist in the Marine Fisheries Department. Explain what you could do to check if the claims made by the fisherman were true. Outline a plan of action to follow if the claims were true. Give reasons for your decisions.
- 6 Give two reasons why fishermen might catch sharks.
- 7 What important role do crayfish play in the marine environment?
- 8 What is meant by a 'non-commercial species'?



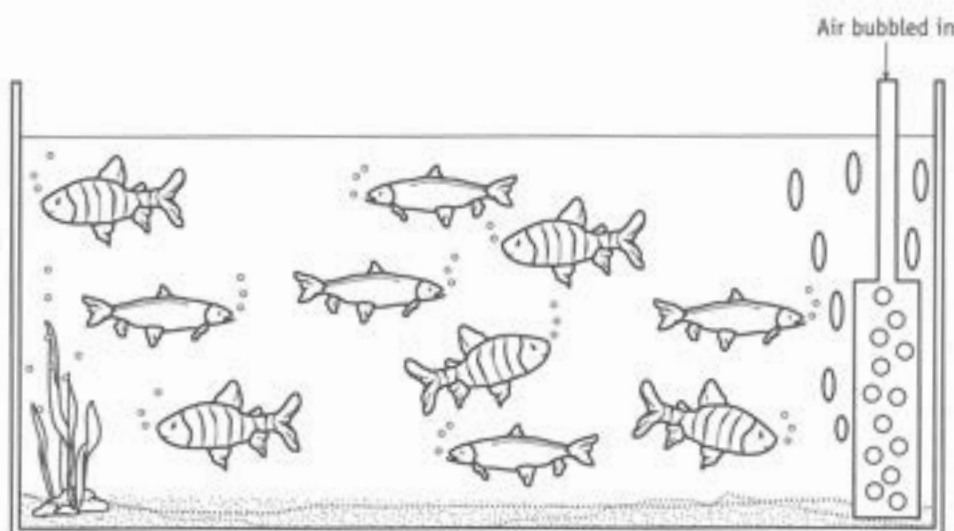
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- 9 What are 'plankton'?
- 10 Give a definition for 'crustacean'.

Survival

All plants and animals depend on one another. Let us suppose two species of organisms are living in the same place and need the same things to survive. It could be two species of animals eating the same food, or two species of plants taking in the same water, carbon dioxide and sunshine. If there is plenty to go round, both types of organisms will survive. If there is not enough, then the two species must fight or compete for the things that they need. Which will survive? The following example might help to decide. A pond was set up in a back garden and stocked with fish.

- There were two species of fish, one plain the other striped.
- Fifty plain fish and fifty striped fish were put in the pond.
- A combination of an aerator and green plants kept the oxygen level high enough to support 100 fish.
- The plain fish were, on average, faster swimmers and more cunning than the striped fish.
- Both species ate the same fish food, but only enough food was put in to support 70 fish.



- 1 For what will the two species of fish need to compete?
- 2 What will gradually happen to the numbers of plain fish?
- 3 What will gradually happen to the numbers of striped fish?
- 4 Give reasons for your answers to Questions 2 and 3.
- 5 Suppose the striped fish could eat the green plants as well as the fish food. What difference, if any, would this make?
- 6 As well as starting off with the same number of plain fish as striped fish, name two other factors to be considered to make this a fair test of the survival capabilities of the two species.
- 7 In real life, for instance in a billabong or creek, both species of fish would be hunted and eaten by predators. Give two examples of likely predators.
- 8 If these fish existed in a billabong or creek, no one would add fish food. What would the fish eat instead?



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- 9 Choose a native Australian species of fish. Give its common name and its scientific name. Describe its habitat and what it eats and/or what eats it. Include one other fact about the fish.
- 10 Define 'billabong' and 'creek'.

The Heart

The cells in your body need food, water and oxygen and must get rid of waste products. A transport system is needed to move these materials through the body. Your blood system does this.

The blood system consists of blood vessels (tubes) of different sizes and a muscular heart which contracts rhythmically to pump blood through the blood vessels. The blood system is often called the *circulatory system*.

The blood vessels that carry blood away from the heart are called *arteries*. Arteries have thick, muscular walls which contract in rhythm with the heart and help force blood along. The regular muscular contractions are your pulse, and you can feel your pulse in your wrist and neck.

The blood vessels that carry the blood back to the heart are called *veins*. Veins have narrower, less muscular walls than arteries and the pressure of the blood is lower. They are found closer to the surface of your skin, and rely on the movement of other muscles in your body to help push blood back to the heart.

The large arteries and veins connected to the heart form numerous branches and become narrower. The narrowest arteries and veins are connected by microscopic vessels called capillaries. Capillaries are very, very thin (only a few cells thick) and allow substances to pass easily through their walls. This is how food, water and dissolved oxygen pass from the blood to the cells, and how waste products from the cells pass into the blood to be carried away.

The human heart weighs about 300 g on average, and is a pump with walls made of muscle. There are specialised tissues in the heart that produce an electrical signal at regular intervals. This signal stimulates rhythmic contractions of the cardiac muscle. When this muscle contracts, it squeezes blood out of the heart and around the body. When the muscle relaxes, the heart fills with blood returning from its journey around the body. One complete contraction and relaxation of the heart makes up a heart beat. Your heart beats about 70 times a minute when you are resting, rising to about 140 times a minute during exercise.

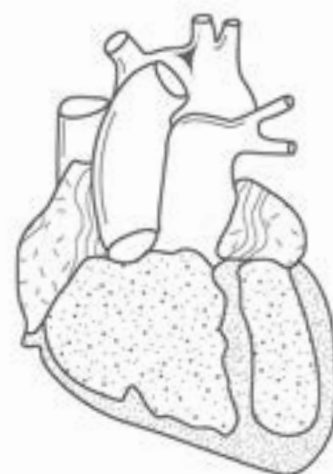
- 1 At 70 beats per minute, 24 hours a day, 365 days a year, how many beats will a heart make in a lifetime of 70 years?
- 2 Why does your pulse rate increase when you exercise?

Sometimes *pacemakers* must be implanted in people who can suffer from major interruptions to the regular rhythm of the heart. These electronic devices maintain a steady heart beat. Pacemakers have to be made of unreactive materials.

The space inside the heart is divided into four compartments. The two top compartments are called *atria* (singular: *atrium*). The atria have thin muscular walls. The two lower compartments are called *ventricles* and have much thicker muscular walls.

- 3 Give a reason why ventricles need to be made of thicker muscles than atria.

Blood can only flow through the heart in one direction. This is because blood flow is controlled by valves. There are valves between the atria and ventricles. These valves are flaps of skin held in place by tendons of tough fibre. They open when blood is pumped forward and shut when blood tries to flow back. Sometimes these valves are faulty and need to be replaced. Teflon™ (polytetrafluoroethylene) is a suitable synthetic replacement material because of its unreactive nature and smooth surface.



The Heart

4 What is meant by 'synthetic'?

Atherosclerosis is a major form of heart disease in Western countries. It is caused by the build up of plaque on coronary arteries and veins. Plaque is made up of cholesterol and fats. If the blood vessels are narrowed by this build up, then blood flow to the heart can be restricted. The build up of plaque may become serious enough to cause a heart attack if it blocks the coronary artery.

Plaque can be reduced and blood flow eased by:

- reduction in dietary intake of animal fats and cholesterol
- increased physical exercise
- drugs
- coronary by-pass surgery.

Heart transplants are a possibility for people with heart disease, but are still controversial and difficult. One major problem is due to the body's immune system rejecting transplanted organs. The body recognises 'foreign' proteins on tissues and treats them as invaders. Christiaan Neethling Barnard was the first person to perform a human heart transplant operation. Artificial hearts have also been constructed, but these are very large and cannot be implanted into the body.

*'It is infinitely better to transplant a heart than to bury it
so it can be devoured by worms' C. Barnard*

- 5 What allows blood to flow in only one direction?
- 6 What do all cells in the body need to survive?
- 7 What are the blood vessels called that take blood away from the heart?
- 8 What are the blood vessels called that take blood to the heart?
- 9 Which blood vessels connect arteries and veins?
- 10 What makes up one complete heart beat?
- 11 What does a human heart weigh, on average?
- 12 What substance makes up the walls of the heart?
- 13 What does this define? 'An electrical device producing a regular electrical impulse to steady the heart beat or reestablishing the rhythm of an arrested heart which has stopped.'
- 14 Name two ways in which a person can reduce the amount of plaque on blood vessels.
- 15 There are many materials and devices that are used to replace diseased or damaged parts of our body. Give two properties needed for a material that is to be used to repair or replace heart valves.
- 16 Why are the walls of capillaries very thin?
- 17 Many people are waiting for heart transplants. Give two reasons why people may not be able to have this operation as soon as it is diagnosed.



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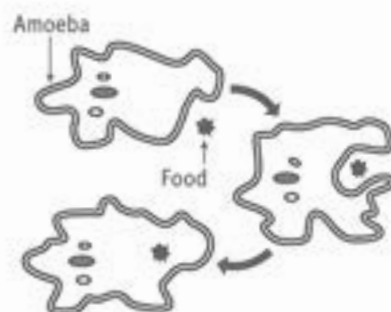
- 18 Angioplasty is the surgical repair of a blood vessel. Explain how it is carried out and how it eases blood flow to and from the heart.
- 19 State some strategies used to minimise the chances of rejection of a transplanted heart.
- 20 Smokers have a greater risk of heart disease than non-smokers. Explain why this is so.
- 21 Some specialists recommend that half an aspirin tablet a day will cut down the risk of a heart attack. Explain the reason behind this.

Respiration

Plants and animals use sugars and oxygen in a process called *respiration* to provide energy for living.



Respiration occurs continuously in living things, during both day and night. For respiration to occur in cells, the sugars and oxygen must reach the cells. Very simple animals, such as amoebas, engulf their food by a process called *phagocytosis* and absorb oxygen from the surrounding water.



More complex animals cannot simply surround their food or absorb oxygen. These animals usually have separate systems working together to make sure that both the oxygen and the food reach the cells. Oxygen must be obtained from the air and transported around the body. Foods need to be broken down so that the cells can absorb the necessary nutrients.

- 1 Explain what is meant by 'nutrient'.

The body's respiratory system is the system that extracts oxygen from the air, uses the oxygen (and food) to supply energy to the cells and returns unwanted gases to the atmosphere. These gases are carbon dioxide, water vapour and unused oxygen.

- 2 Give the chemical formula for carbon dioxide.
- 3 Give the chemical formula for water.



You may have done some chemical tests to identify carbon dioxide, water and oxygen. Carbon dioxide turns a solution called limewater from clear to cloudy. Water turns cobalt chloride paper from blue to pink. Oxygen will relight a smouldering piece of wood.

In the human respiratory system, air is taken in through the nose or mouth. It travels down a long pipe called the *trachea*. The trachea then splits into two *bronchi* which lead to the lungs. The bronchi branch into tiny tubes called *bronchioles* inside each of the lungs. The bronchioles end in bubble-like air sacs called *alveoli*. One alveolus is about 2 mm in diameter. Air inside each alveolus gives oxygen to the blood and receives carbon dioxide from the blood through capillaries (tiny blood vessels). This system gives a very large surface area for gases to exchange between the lungs and the blood stream. Exhaled air contains more carbon dioxide than inhaled air. If you see or hear the word 'pulmonary' in a sentence or phrase, it means that the sentence or phrase has something to do with the lungs.

Air gets to the air sacs by the process of breathing. When you breathe in, the rib-cage moves out and up, and the volume of the chest space increases. This action is brought about by the action of muscles attached to the ribs. The diaphragm is a sheet of muscle attached to the bottom of the ribs. It is not flat but is curved. When it contracts, however, the diaphragm becomes flat. As a result of this, and the movement of ribs, the volume of chest space increases. This draws air into the lungs. When you breathe out, the ribs move down and in. The diaphragm relaxes and returns to its domed shape. The volume of the chest space gets smaller. This forces air out of the lungs.

- 4 Name three things that a human's respiratory system does.
- 5 What material makes up the diaphragm?
- 6 How many bronchi are there in a human body?
- 7 What substances are used up in respiration?

Respiration

- 8 Explain how the ribs move when you inhale.
- 9 Explain how the ribs move when you exhale.
- 10 Describe the chemical test for the gas oxygen.
- 11 Describe the chemical test for the gas carbon dioxide.
- 12 What is the name for the organs that connect the trachea to the lungs?
- 13 What are capillaries?
- 14 What shape are alveoli?
- 15 Which contains more carbon dioxide, inhaled air or exhaled air? Explain your answer.
- 16 Another name for the trachea is the 'wind pipe'. True or false?

Sometimes a person cannot breathe on his or her own and 'artificial respiration' is needed. This involves air being forced in and out of the lungs of the affected person. This can be done by a machine or by another person. Normal breathing can be stopped by heart failure, drowning, electric shocks and so on. Emergency services carry *resuscitators*, devices used to restore breathing. *Cardio-pulmonary resuscitation* (CPR) is a procedure used on a person who has suffered cardiac arrest (his or her heart has stopped). In this, the person administering CPR breathes for the victim and also presses on the victim's chest to keep blood circulating through the body.

Sometimes surgery, for instance 'open-heart', will require heart-lung machines to take over the function of these organs to ensure a patient's survival. Another consideration is that there is a likelihood of brain damage if the brain is deprived of oxygen for more than five minutes.

Hospitals carry a range of life-support systems. These can include devices which supply vital fluids (intravenous drips) and oxygen, keep vital organs going, or monitor vital signs such as pulse and breathing rate.

- 17 What does 'monitor' mean?

Poliomyelitis ('polio') is a (once common) disease which can destroy the nerves responsible for respiration. Patients suffering extreme forms of the disease need to be placed in an 'iron lung'. This device causes air to move in and out of the affected person's lungs without effort from the person.

- 18 Name three things which can cause a person to stop breathing.
- 19 What is the full name for CPR?
- 20 What does a *resuscitator* do?
- 21 Why must the brain receive a constant supply of oxygen?



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- 22 Define 'asphyxiate'.
- 23 Describe two liquids likely to be supplied by intravenous (IV) drips.
- 24 Cardio-pulmonary resuscitation is a method used to restore normal breathing after cardiac arrest. Detail how this technique is used by one person carrying out the procedure.
- 25 Detail how CPR is used by two people carrying out the procedure.

Polio is a disease which was once much more common than it is now.

- 26 Describe the symptoms of polio.
- 27 Explain how an 'iron lung' works.
- 28 Explain why the occurrence of polio is much less now than it once was.
- 29 Draw or photocopy a diagram showing the shapes and relative sizes of bronchioles and alveoli, and how they are connected.

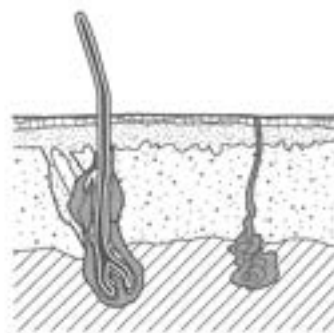
The Skin

The skin (or cutis) is the largest organ of the body. It is a protective covering and holds other organs inside the body. The thin outer part is called the *epidermis*, in which cells multiply continuously and push dead cells outward. Cells (melanocytes) in the epidermis contain a pigment (colouring matter) called *melanin*, which gives the skin its colour.

In the presence of sunlight the skin produces Vitamin D, an essential ingredient for healthy bones and teeth. Although we need to be in the sun for a certain amount of time to maintain health, too much sun is bad for us. It can cause sunburn, dry skin and, over a period of years, cancers can develop. The major cause of this is ultraviolet (UV) rays.

A thicker, inner layer is the *dermis*, or true skin. The dermis contains connective tissue which has blood vessels, nerve endings, sweat glands, oil glands, hair follicles, muscles and fatty tissue. The skin has many sensations, because the nerve endings that can detect touch, heat, cold, pressure and pain are present in it. The ends of nails and hair are dead cells, so it does not hurt to cut them.

Oil produced in the oil (sebaceous) glands help keep the skin smooth and soft. Hair grows from follicles (tiny tubes) that extend into the dermis where cells divide rapidly. The length of the hair depends on the location in the body. For example, eyelashes only grow to a certain length.



Tiny openings or pores are in the skin. These allow us to rid ourselves of waste products through perspiration (sweat). This also helps us to maintain a steady body temperature of 37°C. When your body is cold, muscle fibres on either side of hair follicles contract. This gives you the feeling of 'goose bumps'. When this happens, air is trapped between the hairs. This provides insulation and helps to keep us warm. Underneath the dermis is a layer of fatty (adipose) tissue which acts both as a fat store and an insulating layer. Capillaries are tiny blood vessels which supply food and oxygen to various parts of the skin.

- 1 Give the names of the outer layer and inner layer of skin.
- 2 What is the colouring matter in skin called?
- 3 What can the skin detect or feel?
- 4 What is the normal body temperature of a human?
- 5 Which important vitamin does the skin produce in sunlight?
- 6 Is too much sun good for us? Explain your answer.
- 7 Why doesn't it hurt when you cut your nails?
- 8 What is the name of the glands which produce oil?
- 9 What makes up adipose tissue? What two jobs does it do?
- 10 What are pores?
- 11 What happens when you get 'goose-bumps'?
- 12 A shampoo promises to 'bring back life into your hair'. What is your scientific comment on this?
- 13 What are capillaries?
- 14 Could identical twins have identical fingerprints? Give a reason for your answer.



RESEARCH

- 15 Why might newborn babies in hospital have copies taken of their feet prints?
- 16 How does unprotected exposure to the sun cause skin cancers?
- 17 You may have seen labels such as SPF 15 on a bottle of suncream. What does this mean?

Our Skeleton

Humans are *vertebrates*. This means that we have a backbone. All vertebrates have a *skeleton*. A skeleton is made up of *bones* or *cartilage* (gristle) connected together to form a semi-rigid structure. Sharks and rays have skeletons made of cartilage. The skeleton provides support, allows movement and protects vital organs.

1 Give an example of a 'vital organ'.

Without a skeleton we would not be able to stand upright. It allows the body to move because many of our limbs (such as legs and arms) have joints. The main parts of a human skeleton are the skull, ribs, vertebral column (backbone), humerus, radius, ulna, femur, tibia, fibula, pelvis and scapula. There are 206 bones in the human skeleton.

The human skeleton has components other than bones, such as cartilage. Cartilage is very tough — the end of the nose and the outer parts of the ears are made of cartilage. Joints are where bones or cartilages join together. Ligaments connect the joints of bones together. Tendons (sinews) join muscles to bones.

There are many types of joints. They include fixed joints which allow no movement, such as in the skull. The bones of joints which move are covered in cartilage (which is smooth) and have a lubricant called synovial fluid.

2 What does a 'lubricant' do?

Ball-and-socket joints (for instance the hip joint) are the most flexible joints. The movable bone can swivel in many directions. Hinge joints (such as the knee) only allow the bone to move in one of two opposing directions. Pivot joints only allow rotation. These are found in the elbow (between the bottom of the humerus and the radius). Sliding (gliding) joints, are found in the wrist and ankle. In these joints the surfaces of the bones slip a short distance over each other.

Sometimes joints suffer 'wear and tear' and need to be dealt with by surgery. Silicones are useful materials for some body-replacement parts. They resist acids, are flexible and are impervious to water.

3 What is meant by 'impervious to water'?

Silicones are particularly suitable for use in small toe and finger joints. They are not suitable for joints which bear large forces.

Ultra-high molecular weight polythene (UHMWPE) can be used to replace worn cartilage around a ball-and-socket joint. It is durable, has low friction and does not react with the surrounding tissue.

4 What does 'durable' mean?

When bone must be replaced, strong alloys need to be used. They must be light and chemically inert.

5 What is meant by 'chemically inert'?

6 What function is served by ligaments?

7 Where is synovial fluid found?

8 Why are silicones suitable materials to use for joints which bear large forces?

9 Draw up a table of the types of joints mentioned on this sheet. Research how these joints work and where they are found in the body.

10 What properties make UHMWPE a good replacement for cartilage?

11 What joins muscles to bones?



RESEARCH

12 Define 'bioengineering'.

13 Define 'alloy'.

14 Define 'silicone'.

Osteoporosis

Read the following (abridged) news article and answer the questions.

Osteoporosis test — now available locally

An Australian chain of rural pharmacists announced that they had secured the exclusive rights to use a new device to test for osteoporosis. This is a condition where, due to a decrease in mass, bones become more porous and more easily fractured than normal. No longer do people need to go to a big city hospital to see if they are at risk.

'We have ordered a new screening tool for all our major outlets. The clinical bone sonometer quickly measures bone density in a fraction of the time, cost and risk of conventional methods', claimed a spokesperson for the company. 'Previously the only way to test for osteoporosis was to have an X-ray

taken using very specialised and expensive equipment. This can cost hundreds of dollars and involve travel, which can be a problem for old and frail people. Our machines are small, inexpensive and portable. They don't require specialist technicians to operate them. As a result, we anticipate to charge a fraction of fees charged previously by large medical facilities and have results ready in minutes.'

The safety factor was also emphasised. 'Because X-rays aren't used, it poses very little hazard. The machine uses ultrasound — sound waves you can't hear. All you need to do is put a bare foot in the machine and for

about ten seconds, sound waves painlessly pass through your foot. The denser the bone, the more sound is affected as it passes through. The denser the bone — the healthier it is, as simple as that.'

A spokesperson from a city hospital sounded a note of caution. 'The new device is a useful screening tool, but isn't a definitive test. You need to use the results in conjunction with your doctor's advice to decide if you need to use the conventional, and much more accurate, dual energy absorptiometry test.' Advice was also given on how to avoid the disease. 'We strongly recommend a high calcium intake and weight-bearing exercise for those groups at risk.'

- 1 What are the symptoms of osteoporosis?
- 2 What is the connection between bone density and bone health?
- 3 Name three benefits claimed by the pharmacy chain for their new device.
- 4 What does the new machine use to test bone density?
- 5 Name three reasons why the new test should be cheaper than conventional X-ray pictures.
- 6 Why might people be concerned about having X-rays taken?

In people suffering from osteoporosis, the most common fractures are those of the hip, wrist and spine. The ultrasound test checks the bone density of a person's heel.

- 7 What assumption is being made in using the bone density at the heel to act as an indicator of osteoporosis?
- 8 Is the clinical bone sonometer likely to be 100% effective in detecting osteoporosis? Explain your answer.
- 9 Give two ways in which people at risk may avoid or minimise the effects of osteoporosis.



RESEARCH

- 10 Find out the types of people who are likely to suffer from osteoporosis.
- 11 Give three causes of osteoporosis.
- 12 Give two ways to treat osteoporosis not already mentioned.

Lifestyle Chemistry: Assignment

NAME _____ DUE DATE _____

In this assignment you are expected to utilise a range of resources. You should include research from most of the following sources:

- dictionary
- text book
- periodical (such as magazine, newspaper)
- CD-Rom
- encyclopedia
- the Internet.

Credit will not be given for work that has been copied directly from a published source. Include a bibliography.

Give brief (two or three lines) definitions for your answers to Questions 1 to 5:

- 1 Define 'colloid'.
- 2 Define 'particle'.
- 3 Define 'lubricant'.
- 4 Define 'pesticide'.
- 5 Define 'solvent'. (1 mark each)



Here is a dictionary definition of an emulsion: 'A system made up of a liquid dispersed in an immiscible liquid usually in droplets of larger than colloidal size' (for instance, fat in milk). Emulsions and emulsifying agents are commonly used in cleaning products.

- 6 Use your own words to describe what an emulsion is. (2 marks)
- 7 Explain, in your own words, what an emulsifying agent does. (2 marks)
- 8 Give an example of an emulsifying agent. (1 mark)

Some manufacturers of skin and hair products mention pH in the advertisements for their products. The pH can be described as a measurement of acidity.

- 9 Explain what the pH scale is and how it is used to measure acidity. (2 marks)
- 10 Find one product which mentions pH. You can cut out a newspaper/magazine advertisement or describe one which is on TV. Comment on how it refers to pH. (2 marks)
- 11 Why might companies want you to know the pH levels of their products? (2 marks)
- 12 Choose a particular brand of hygiene product such as soap, skin cleanser, or shampoo. Name the product and list the ingredients. (1 mark)
- 13 Pick two of the ingredients you listed in Question 12. Explain what they do — that is, why they are used in making that product. (2 marks)

People take medicine for many reasons. Medication comes in many forms.

- 14 Find three different ways in which medicinal drugs are taken. (3 marks)
- 15 For each example in Question 14 explain why that particular form is used. (3 marks)
- 16 Why are artificial flavours and colours sometimes added to medicines? (1 mark)

Solubility of Tablets: Practical assessment

Many manufacturers of pharmaceuticals often promote as a selling point the quick dissolving properties of their tablets and capsules, particularly headache remedies. Other manufacturers stress products which are 'slow release', taking longer to dissolve and allowing the medication to be absorbed gradually by the consumer.



**Student note: do not consume tablets or mixtures.
Wash hands after working with tablets and mixtures.**

You need to devise and then carry out an experiment to compare how quickly different tablets dissolve in water. You may use any common laboratory apparatus.

First explain how you intend to carry out the experiment. In this you must include:

- how you could tell if the tablet had dissolved
- what conditions you would vary
- what apparatus you would use
- how you would ensure that it was a fair test
- what precautions you would take when carrying out the experiment
- why it might be beneficial for medication to be soluble.

When you have done this, show your work to your teacher *before* carrying out the practical experiment. Do *not* attempt any practical work unless it has been approved by the teacher. **You are reminded of the necessity to observe appropriate safety precautions at all times.**

In your final write up of the practical experiment you carried out, you should include:

- a clear method indicating what changes you made to your original plans. (Explain why any changes to the original method were made.)
- the type of tablet used
- a clear table of results, including quantities and units where appropriate
- a labelled diagram of the apparatus used
- a conclusion — this should include how you interpreted your results and any problems you faced or improvements that could be made to future experiments.

Note to teacher: as with any practical work, it is necessary to carry out a risk assessment before carrying out this activity and to observe all necessary precautions.

Astronomy

A comet is a misty celestial object revolving around the sun in a huge elliptical orbit.

- 1 What is meant by a 'celestial object'?

It has a small, solid nucleus with a long tail which points away from the sun. The most famous comet is Halley's Comet which came close to Earth in 1531, 1607, 1682, 1759 and so on. with its latest appearance being in 1986.

- 2 When do you think Halley's Comet came near the Earth between 1758 and 1986? Explain your answer.

Nebulae are clouds of gas and dust particles. Before telescopes were used in astronomy, many star groupings were mistakenly called nebulae because of their diffuse appearance. A famous dark nebula is the Horsehead Nebula in Orion.

- 3 What is meant by 'diffuse appearance'?

A nova (plural: novae) is literally a 'new star'. It is actually an existing star that gets many thousands of times brighter than normal without warning and then slowly fades back to its original brightness. A supernova (plural: supernovae) is a star which increases its brightness billions of times and the star is destroyed in the process. Supernovae are much rarer than novae.

- 4 What would happen to life on Earth if our sun became a nova?

All stars give out radiation including visible light. Astronomers analyse the light from them with a spectroscope. This splits up light into a 'rainbow' spectrum of colours. Specific lines in the spectrum of an object can tell astronomers the materials of which the object is made. An object which is moving rapidly away from an observer has its spectrum 'shifted' towards the longer wavelengths, that is, red. An object which is moving rapidly towards an observer has its spectrum 'shifted' towards the shorter wavelengths, such as blue. Astronomers have discovered that the spectra from other galaxies has displayed 'red shift', the further away galaxies showing the greatest 'red shift'.

- 5 What do you think astronomers inferred from this 'red shift' of galaxies?

Artificial satellites are now being used to detect radiation which is blocked or distorted by the Earth's atmosphere. These include IRAS (the Infra Red Astronomy Satellite) and the Hubble Space Telescope.

- 6 What does this define? 'A massive self-luminous, gaseous sphere of great mass which produces energy by means of nuclear fusion reactions.'
7 Is a comet a satellite? Explain your answer.
8 Explain the difference between a 'nova' and a 'supernova'.



Answer true or false to each of the following.

- 9 Stars only give out visible light. _____
10 Satellites can get better pictures of the universe than Earth-based observatories. _____
11 A nebula is a solid object. _____



RESEARCH

- 12 There are many theories of how the universe was formed. Most scientists believe in either the 'Steady state' model or the 'Big bang' model. Briefly explain what lies behind each of these theories.
13 The Hubble Telescope is named after Edwin Hubble. For what is he famous?
14 Find out two facts about Sir Edmund Halley.

The Development of Astronomy

Astronomy: the study of planets, stars and other heavenly bodies.

Humans have always been interested in the skies. People have been frightened by comets, meteor showers, eclipses of the sun and of the moon and so on. Many ancient civilisations came up with calendars based on astronomical investigations, which they used to work out when it was time for planting, harvesting and so on. A summary of advances in astronomy is given below.

- Almost 2000 years ago, an Egyptian astronomer named Claudius Ptolemy said the Earth was the centre of the universe. He also stated that all objects in the sky were perfect spheres. For the next 13 centuries this idea (the Geocentric theory) was held by most people.
- A Polish monk, Nicolaus Copernicus (around 1530), came up with the idea that the sun, and not the Earth, was the centre of our solar system. This suggestion (the Heliocentric theory) disagreed with the beliefs of the Church at the time. His work was not published until just before his death in 1543. It took a long time and the work of many scientists to prove that Copernicus was right.
- Tycho Brahe (1546 to 1601) was a Danish astronomer who, with the assistance of his sister Sofie, gathered very accurate astronomical observations. He didn't agree with all of the theories of Copernicus. His work was used by his German assistant, Johannes Kepler (1571 to 1630), to prove planets did not move in circular orbits, but in ellipses (ovals).
- Galileo Galilei (1564 to 1642) is credited with being the first person to use the telescope to make astronomical observations. He saw the Earth's moon had craters, Saturn had rings and that there were moons orbiting Jupiter. The idea that other planets had satellites, contradicted the idea that the Earth was the only object that had others orbiting it. Like Copernicus, Galileo's ideas challenged currently held beliefs and were not liked by some powerful people at the time. He was put under confinement from 1638 to 1642.
- Sir Isaac Newton (1642 to 1727) used mathematics and the concept of gravity to give reasons for the ways heavenly bodies moved in orbits.



To navigate, sailors once used observations of the sky to work out the time and position. The invention of *sextants* and *chronometers* improved their accuracy. Sextants were instruments using celestial bodies to measure angular distances. Chronometers are very accurate clocks.

- 1 Give your own definition of 'astronomy'.
- 2 Briefly outline the difference between the Geocentric theory and the Heliocentric theory.
- 3 What is a 'comet'?
- 4 Who was the first person to suggest that the sun was the centre of our solar system?
- 5 Why did ancient civilisations look at the heavens to make calendars?
- 6 What does the word 'lunar' mean?
- 7 What did Galileo see through his telescope?
- 8 Draw up a time line showing the dates of the lives, and the discoveries, of the European astronomers mentioned.



RESEARCH

- 9 Find out how the first chronometer was developed by the horologist (clockmaker) John Harrison (1693–1776) and why this was so important.
- 10 Explain the difference between 'latitude' and 'longitude'.
- 11 Find out how our modern calendar has developed. (Hint: Babylonian, Egyptian, Roman, Gregorian.)

Astronomy and Australia

The view of stars and planets in the Southern hemisphere is different from the view in the Northern hemisphere. In 1768, in command of the *Endeavour*, James Cook was sent to the South Pacific. He, with a group of British astronomers, observed the transit of the planet Venus across the sun in June 1769. Following this he charted the coasts of New Zealand, Australia and New Guinea.

The Astronomer Royal in England (Nevil Maskeleyne) expected a comet to reappear in 1789 and wanted an observatory set up in the Southern hemisphere. He appointed Lieutenant William Dawes to travel with the First Fleet to Australia. Dawes constructed his observatory in Sydney (at Maskeleyne Point, now known as Dawes Point) but the comet never appeared. His son (William Rutter Dawes) became a well-known astronomer.

Sir Thomas Brisbane was a British soldier and astronomer who was appointed Governor of New South Wales in 1821. His interest in astronomy began when he was nearly shipwrecked due to a navigational error. He established an observatory in 1822 at Parramatta, near Sydney. He appointed two astronomers to study the stars visible in the Southern hemisphere and, by 1835, they had described and classified thousands of new stars.

Australia has had its share of famous astronomers. John Tebbutt was an amateur astronomer born in Windsor, NSW. He started his observations in the 1850s and built his first observatory in 1863. He was the most famous Australian astronomer of his time. He discovered two comets and was also interested in meteorology (the atmosphere and weather).

Another notable Australian astronomer was Joseph Lade Pawsey who died in 1962. He specialised in radioastronomy and worked for the CSIRO.

Celestial objects often give out radio waves. These are similar to light, but are of a longer wavelength and cannot be seen. They can be detected by radio telescopes. Radio telescopes are very large, dish-shaped reflectors which collect radio waves and focus them onto a receiver.

A number of observatories were built throughout Australia during the nineteenth century. There are many places where the air is clean and there is no 'light pollution', such as city lights, to interfere with astronomical observations. Today we have many optical and radio telescopes. Astronomers from all over the world use the radio telescope at Parkes (NSW). It has a steerable dish with a 64 metre diameter.

- 1 Why did Captain Cook come to the Southern hemisphere?
- 2 What is an 'observatory'?
- 3 What is 'meteorology'?
- 4 Why might astronomers site their telescopes far away from cities?
- 5 What is a 'radio telescope'?
- 6 Why did Lieutenant William Dawes come to Australia?
- 7 Give details of two contributions that Sir Thomas Brisbane made to Australian astronomy.
- 8 An astronomer studies celestial objects. True or false?



RESEARCH

- 9 Which Australian astronomer once had his picture on our \$100 note?
- 10 What is a 'planetarium'?
- 11 Name one well-known Australian astronomer not mentioned previously.

Astronomy: Assignment

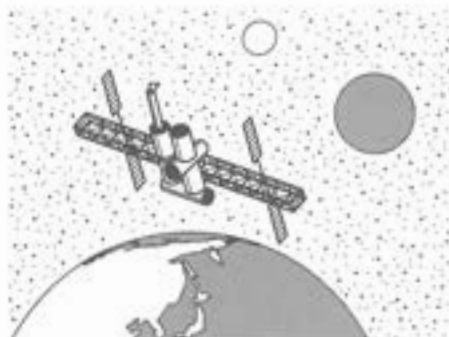
NAME _____ DUE DATE _____

Part A: Identify the following astronomers using the descriptions they give of themselves. Then find out one *additional* fact about each person.

Choose from these astronomers:

Isaac Newton	Edmond Halley	Galileo Galilei
Johannes Kepler	Copernicus	Tycho Brahe
Ptolemy	Edwin Hubble	Plato

(Note: not all of these names are used.)



- 1** I lived in England from 1656 to 1742. As an astronomer, I observed the comet of 1682 and correctly predicted its return in 1758. This comet now bears my name.

Who am I? _____

Fact _____

- 2** I was born in Poland and have often been credited with being the pioneer of modern astronomy. I lived from 1473 to 1543. I felt sure that the sun was the centre of the universe and that the planets moved around it in circles.

Who am I? _____

Fact _____

- 3** I lived in Alexandria in the second century AD. I believed that the Earth was the centre of the universe and that not all objects in the skies were perfect spheres.

Who am I? _____

Fact _____

- 4** I lived from 1642 to 1727. Although famous for my Law of Gravitation, I also worked with light and prisms. I made a telescope that worked with mirrors.

Who am I? _____

Fact _____

- 5** I was an American astronomer who lived from 1889 to 1953. I have had a modern telescope named after me.

Who am I? _____

Fact _____

Astronomy: Assignment

Part B: The descriptions of some of the planets of the solar system have been put in a table. Write the name of the correct planet next to its description.

Choose from:

Earth

Jupiter

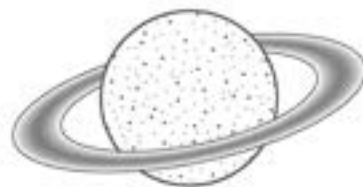
Mars

Neptune

Venus

Mercury

Saturn



Planet	Description	Name
6	Named after the 'God of the Sea'. Placed between Pluto and Uranus.	
7	Has phases like the moon. Covered in clouds — a very bright object in the sky.	
8	Only planet known to have life on it. 'Third Rock From The Sun'.	
9	Largest planet, also has a 'red spot'.	
10	Known as the 'Red Planet'. Has moons called Phobos and Deimos.	
11	Has bright rings and many moons, including one called Titan.	
12	Closest planet to the sun. Has a metal named after it.	

13 Choose one of the planets above and state three more features not already mentioned.

Part C: Give very brief definitions of the following.

14 Define 'meteor'. _____

15 Define 'satellite'. _____

16 Define 'comet'. _____

17 Define 'constellation'. _____

18 Define 'galaxy'. _____

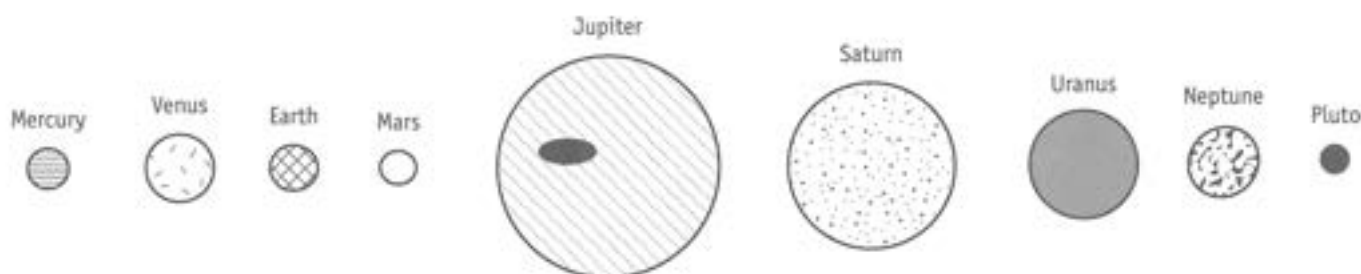


BIBLIOGRAPHY (books and so on used):

The Solar System: Assignment

NAME _____ DUE DATE _____

Below is a diagram of the solar system. It is not to scale. Study it and answer the questions.



- 1 How many planets are in the solar system? _____
- 2 What major object is not shown in the diagram? _____
- 3 Which planet has life present? _____
- 4 Which planet is closest to the sun? _____
- 5 Which planet is farthest from the sun. _____
- 6 Which planet has a 'red spot'? _____
- 7 Saturn has rings around it. Draw these rings on the diagram.
- 8 Why isn't the diagram drawn to scale? _____



RESEARCH

Find out the diameters of these planets:

- | | | |
|-----------------|------------------|-----------------|
| 9 Mercury _____ | 10 Venus _____ | 11 Earth _____ |
| 12 Mars _____ | 13 Jupiter _____ | 14 Saturn _____ |
| 15 Uranus _____ | 16 Neptune _____ | 17 Pluto _____ |

The Solar System: Assignment

For Questions 18 to 22 use the 'scrambled' words in brackets as clues.

- 18 Which planet is sometimes called the 'Red Planet'? (sram) _____
- 19 Which planet is named after the 'God of the Sea'? (penunet) _____
- 20 Which planet has the same name as a liquid metal? (curryme) _____
- 21 Which planet has phases like the moon? (sunve) _____
- 22 Which planet has the largest orbit around the sun? (toplu) _____
- 23 Which object in the solar system are we talking about when we say 'solar'? _____
- 24 When we say 'Martian', to which planet do we refer? _____
- 25 When we say 'lunar' we are talking about a satellite of the Earth called the _____
- 26 Scientists think that the planets of the solar system were formed thousands of years ago. True or false?

- 27 What does this describe? *'The star which, due to its gravitational effect and radiated energy, dominates the solar system'* _____



RESEARCH

Find out the average distance of each of these planets from the sun.

- | | | |
|------------------|------------------|-----------------|
| 28 Mercury _____ | 29 Venus _____ | 30 Earth _____ |
| 31 Mars _____ | 32 Jupiter _____ | 33 Saturn _____ |
| 34 Uranus _____ | 35 Neptune _____ | 36 Pluto _____ |
- 37 Distances in astronomy are often measured in *light years*. What is a 'light year' and why is it used?

- 38 What does this describe? *'The region of interplanetary space between the orbits of Mars and Jupiter in which most asteroids are found.'* _____

Telescopes: Assignment

NAME _____ DUE DATE _____

Astronomers use telescopes to examine the universe. Optical telescopes come in two sorts (*reflecting* and *refracting*), but both use light to observe the heavens. The invention of the telescope is usually credited to Hans Lippershey (a Dutch optician) in 1608. Early telescopes were 'refracting' types.

- 1 Describe how the refracting telescope works _____

Galileo Galilei is credited as being the first person to use a telescope to make astronomical observations.

- 2 Name two things Galileo observed. _____

- 3 When and where did Galileo live? _____

Some of the optical problems associated with making lenses were overcome by using mirrors to collect and focus light. Sir Isaac Newton made the first reflecting telescope in 1668. All large, modern optical telescopes are of the 'reflecting' type.

- 4 Describe how the reflecting telescope works. _____

- 5 Explain why reflecting telescopes are used by astronomers instead of refracting telescopes. _____

- 6 Why are optical telescopes often located near the tops of high mountains? _____

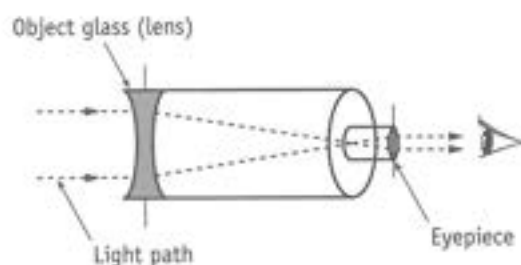
- 7 Explain briefly how radio telescopes differ from optical telescopes. _____

- 8 Give one advantage that radio telescopes have over optical telescopes. _____

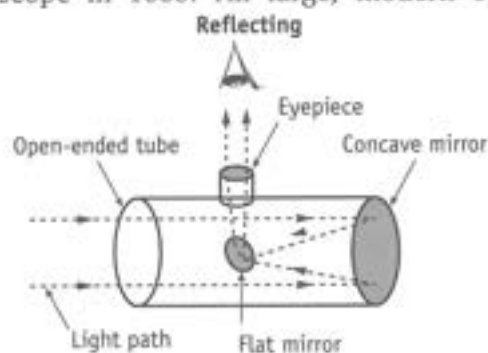


RESEARCH

- 9 Quasars can be detected by radio but not optical telescopes. Give two other facts about quasars. _____



Refracting



Reflecting

Astronomy: Cloze

Use the following words to fill the spaces.

<i>asteroids</i>	<i>astronomers</i>	<i>astronomical</i>	<i>comets</i>	<i>constellation</i>	<i>Earth</i>	<i>elements</i>	<i>gas</i>
<i>light</i>	<i>luminosity</i>	<i>Mercury</i>	<i>meteors</i>	<i>meteoroids</i>	<i>nebula</i>	<i>optical</i>	<i>planets</i>
<i>Pluto</i>	<i>radio</i>	<i>reflected</i>	<i>rotation</i>	<i>satellite</i>	<i>seasons</i>	<i>solid</i>	<i>stars</i>
<i>sun</i>	<i>telescopes</i>	<i>tides</i>	<i>tilted</i>	<i>years</i>			

The Earth is one of nine _____ which orbit the Sun, to make up our solar system. _____ is the planet nearest the sun and _____ is the furthest away. Some planets (such as Mercury, Venus, Earth and Mars) are _____, and some (Jupiter, Saturn, Uranus and Neptune) are _____ giants.

A collection of _____ is called a galaxy. Our galaxy is called the Milky Way. Stars radiate _____, planets only give off _____ light from stars. A group of stars which forms an imaginary pattern is a _____. A _____ is a cloud of gas or dust in interstellar space.

Distances in the universe are so enormous that we often use light _____ to measure them, that is, the distance travelled by light in one year. People who study the stars and other celestial bodies are called _____ and use _____ to investigate the universe. Telescopes can be _____ telescopes, which use lenses and light, or _____ telescopes which pick up the radio signals given out by far distant objects. _____ are rocks and dust particles travelling in space at high speeds. If they enter the Earth's atmosphere, they burn up and are called 'shooting stars' or _____. _____ are celestial bodies made of ices and dust orbiting the sun, with long tails pointing away from the sun. The _____ is a star. _____ are small celestial bodies found between the orbits of Mars and Jupiter.

The _____ moves around the sun once every 365 days (a year) and spins on its axis, once every 24 hours. Night and day are caused by the _____ of the Earth. The axis is _____ over. This tilt of the Earth's axis is the cause of the _____. The moon is a _____ of the Earth and it causes _____ in our seas and oceans. One _____ unit (AU) is the average distance between the Earth and the sun (149 600 000 km). The sun is a typical star with average size and _____ (brightness). The sun is mainly composed of the _____ hydrogen and helium.

Structure of Plants

When scientists talk about the structure of plants, they are usually referring to the plant's roots, stem, leaves and flowers. Not every plant has all of these parts. We are going to look at flowering plants.

The roots anchor the plant to the earth. They help the plant extract water and minerals from the soil. Water flows up from the roots to the leaves (where some evaporates) by a process called *transpiration*. The stem helps support the plant and keep it upright. Liquids travel up and down the stem. In the stem are special tubes called *xylem* which carry water and dissolved salts from the roots to the leaves in the process of transpiration. *Phloem* are tubes which carry sugars (made in photosynthesis) from the leaves to other parts of the plant. Cellulose is a complex carbohydrate and makes up the majority of the cell wall of plant cells.

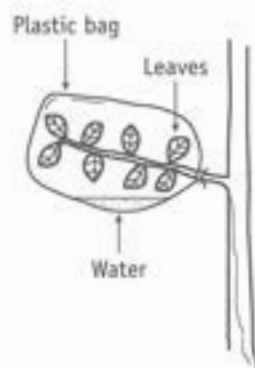
The leaves are where photosynthesis takes place. There is a green pigment (colouring) in leaves called *chlorophyll*, which is essential for photosynthesis. A plant is a *producer organism*. This means that it manufactures its own food. During the day it *respires* (uses up food and generates carbon dioxide) and *photosynthesises* (absorbs carbon dioxide) at the same time. During the night it just *respires*. Animals cannot make food. Animals simply use food made by plants or eat other animals and, in the process, give out carbon dioxide. Carbon dioxide (CO_2) is a colourless, odourless gas which makes up less than 1% of air. Plants help to regulate the amount of carbon dioxide in the air and, hence, affect the greenhouse effect and the world's climate. To enter the plant, carbon dioxide must pass through tiny holes called *stomates* which are found in the largest number on the underside of leaves. They allow oxygen and carbon dioxide to move in and out of the plant. Water vapour can also move out of the plant through stomates.

As part of photosynthesis, plants change the sugars that they make and store them as starch. Iodine solution is a test for starch. When iodine solution is added to starch a blue-black colour results. Only where there is chlorophyll in the leaf, can photosynthesis take place. Variegated leaves are not green all over. ('Variegated' means having different colours.) If tested with iodine variegated leaves only show the presence of starch in the parts which are green (the parts that have chlorophyll present). If leaves are kept in the dark and tested with iodine, they don't show the presence of starch.

Phototropism is the scientific word for saying plants grow towards the light. This is because they need sunlight to provide the energy for photosynthesis.

Flowers are part of the reproductive systems of plants. They contain the sex cells which help to make new plants.

- 1 What is the function of flowers on plants?
- 2 Where does photosynthesis take place in a plant?
- 3 What is the test for starch?
- 4 What two jobs do roots do for a plant?
- 5 How do gases get in and out of leaves?
- 6 A student placed a clear plastic bag over some leaves and left it for a few days. Water collected in the bag although it hadn't rained. Explain why this happened.
- 7 What is 'phototropism'?
- 8 What is a 'producer organism'?
- 9 What happens when a plant *respires*?
- 10 'Xylem' and 'phloem' are tubes which are involved in the transport system of plants. Explain how they are different to each other.
- 11 'Animals can make their own food.' Is this statement true or false?



RESEARCH

- 12 Draw and label a diagram of a typical plant cell.
- 13 Define 'carbohydrate'.

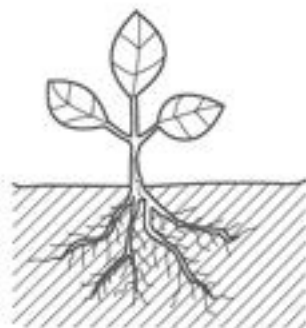
Seeds

Seed: the fertilised ovule of a flowering plant which is able to germinate and produce a new plant.

Many plants begin their lives as seeds. A seed contains a store of food, an embryo and a seed coat (testa). The embryo has the material needed to form the shoot and root. The seed coat protects the seed. When conditions are right, the seed grows — a process called germination — and a seedling is formed. Water, oxygen and warmth are needed before germination can take place.

A root comes from the embryo and grows downwards to anchor the young plant in the soil. At the same time, a shoot grows upwards into the air. Leaves grow on the shoot and the seedling is able to support itself.

Until this stage is reached, the seedling depends entirely on its own food store. The stored food is used up as the seedling grows.



Seeds contain very little water and can absorb more than their own weight in water. Sometimes the seed coat must be cracked to facilitate the absorption of water. Some seeds need to be soaked in water. Others, such as the carob (locust bean), need to have been eaten and passed through an animal's digestive system. The seed must also have a supply of oxygen. Many seeds will fail to germinate if they are lying very deep in the soil. This is because oxygen from the air only reaches them very slowly. All plants have a fairly narrow range of temperatures within which they will germinate. For wheat seeds, the best temperature for germination is between 25°C and 30°C and for maize (corn) seeds it is between 37°C and 44°C.

Some seeds are said to be dormant. The seeds are alive but waiting for the best conditions to germinate. Dormant seeds are able to stay alive even under severe weather conditions which would kill the seedling if germination had taken place.

Most seeds can remain viable (capable of growing) for very long periods. Most seeds used by farmers and gardeners, if stored correctly, can be kept for a year without losing their ability to germinate. The percentage of seeds which will germinate decreases with the length of time they are kept dormant. Eventually, after a long period of dormancy, all seeds will die. It is claimed that seeds over a thousand years old have successfully been germinated and grown into plants.

- 1 Name the three parts of a seed.
- 2 What are the three basic things a seed needs to germinate?
- 3 Are *dormant* seeds dead because they don't germinate straight away?
- 4 Many firms who sell seeds, dry them and then seal them in packets of aluminium foil. On these packets there will often be instructions telling you to store the seeds in a cool, dry place. Why are these measures taken?
- 5 What does the *testa* do?
- 6 You buy a packet of seeds, and carefully read and follow the instructions exactly on how to plant the seeds. Eventually, however, only just half of the seeds germinate. Give a reason why this might have happened.
- 7 Name one region of Australia that might be suitable for growing maize. Give a reason for your choice.
- 8 Why might a carob seed need to have been partially digested before it can germinate?
- 9 What is the difference between a 'dormant' seed and a 'non-viable' seed?

Plants: Assignment

NAME _____ DUE DATE _____

'WHAT AM I?'

Choose from these:

algae conifer eucalypt fern fungus herbivore

- 1 Some people think I am a plant, but I don't make my own food. I reproduce by producing spores. What am I? _____
- 2 I grow cones containing seeds and have thin leaves which I keep all the year round. What am I? _____
- 3 I am a simple form of life but can photosynthesise and have been around for millions of years. What am I? _____
- 4 I am not a plant, but eat plants. What am I? _____
- 5 I smell a bit and my bark is sometimes ragged. I often grow very tall and my leaves hang vertically facing the sun. What am I? _____
- 6 I can be very small or grow to more than 20 m. I have fronds and produce spores. What am I? _____

DEFINITIONS

- 7 Define 'flora'. _____
- 8 Define 'fauna'. _____
- 9 Define 'fruit'. _____
- 10 Define 'vegetable'. _____



RESEARCH

- 11 Is a tomato a fruit or a vegetable? Justify your answer.

- 12 Choose a plant which has not already been mentioned. Find out *three* facts about this plant.

- 13 What is 'geotropism'? _____



Plant Growth and van Helmont's Experiment

People wondered for a long period in history how a tiny seed could grow into a large plant. They were curious as to where all of the extra mass came from. Greek philosophers such as Aristotle (384–322 BC), believed that plants gained their nourishment from the soil.

1 How long did Aristotle live?

It wasn't until the seventeenth century that experiments were carried out to test the validity of this theory. A Belgian physician, Jean-Baptiste van Helmont (1580–1644), demonstrated that soil contributes very little to the mass gained by plants as they grow. He placed a willow seedling of known mass into a pot which contained a known mass of soil. He watered the plant regularly and reweighed the plant and soil after five years. Here is a simplified version of the results of his experiment:

• mass of soil	90.9 kg
• mass of soil plus seedling	91.0 kg
• mass of soil plus plant after five years	167.6 kg
• mass of plant	76.8 kg

2 What was the mass of the willow seedling?

3 What was the change in mass of the soil?

4 Give a conclusion that could be drawn from these results.

Plants extract vital minerals from the soil, but these make up only a very small part of the mass of a plant. Only a small amount of the water taken up by a plant is kept by the plant. Typically, over 90% of water is lost in transpiration. This is the process by which water evaporates from plant leaves. A considerable amount of the mass of a plant is gained from carbon dioxide absorbed from the air.

5 Give the chemical formula for carbon dioxide.

Some elements are required by plants in relatively large amounts and these are called *macronutrients*. The most common of these elements are carbon, hydrogen, oxygen and nitrogen. Others include sulfur, phosphorus, calcium, potassium and magnesium. Other elements needed by plants are iron, chlorine, copper, manganese, zinc, molybdenum, boron and nickel. These are called *micronutrients* or *trace elements*.

6 Define 'micronutrients'.

Hydroponics: the cultivation of plants in a nutrient solution without soil.

7 List the names of the elements mentioned in the passage above and give their chemical symbols.

Hydroponics is a technique which can be used to grow plants in greenhouses out of season, or in areas and climates where these plants cannot normally be grown.

8 Name one food crop grown hydroponically.

Hydroponics is more expensive than conventional farming but the nutrients can be precisely controlled. Hydroponics can be used to investigate the specific needs of different plants.

9 Give two reasons why a farmer might decide to grow a crop using hydroponics rather than the usual method of planting in soil.



RESEARCH

10 Give two facts about hydroponics not previously mentioned.

11 What is the difference between a 'philosopher' and a 'scientist'?

12 The results quoted for van Helmont's experiment use kilograms as units of mass. Would he have used these units? Explain your answer.

13 Find out two facts about Jean-Baptiste van Helmont not previously mentioned.

Growing Plants: Cloze

Use the following list of words to fill in the spaces below:

<i>animals</i>	<i>carbohydrates</i>	<i>cloning</i>	<i>cuttings</i>	<i>dispersed</i>	<i>dormant</i>	<i>flowers</i>	<i>food</i>
<i>genetically</i>	<i>grafting</i>	<i>human</i>	<i>identical</i>	<i>insects</i>	<i>leaves</i>	<i>ovary</i>	<i>photosynthesis</i>
<i>planting</i>	<i>pollination</i>	<i>seeds</i>	<i>self</i>	<i>stamens</i>	<i>viable</i>	<i>water</i>	<i>warmth</i>

Seeds can remain _____ (inactive) for many years. Seeds which can be made to grow are said to be _____. When a seed is placed in damp soil, it immediately starts to absorb _____. If oxygen and _____ are also provided, the seed will germinate. Approximately seven days after _____, shoots of the germinating seeds show through the soil. The two _____ at the top of the shoot are called cotyledons, these originally contained the seed's store of _____. Once the seedling has used up its store of food then it obtains energy from the sun by a process called _____. _____ leaves contain chlorophyll which helps the plant to change carbon dioxide and water into _____.

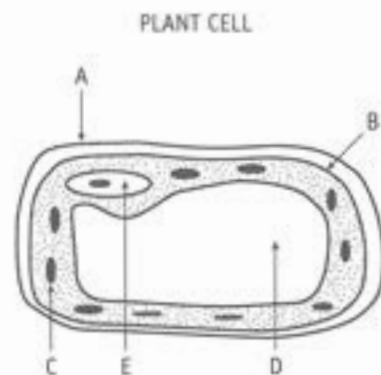
_____ are the reproductive organs of plants. Eventually flowers develop into the fruit and _____. Before this can happen _____ must take place. This is the transfer of pollen from the _____ to the stigma in the centre of the flower. Sexual reproduction means that the plants formed are not _____. Pollination can be carried out by _____ or the wind, but some plants (such as tomatoes) can _____-pollinate. If fertilisation occurs, the petals of the flower wither and seeds develop inside the _____ which expands to form the fruit. Fruit is a means by which seeds are _____. _____ eat the fruit. The seeds pass through the animal's gut. Seeds can also be collected by _____ to grow new plants.

_____ is a process where a portion of one plant is placed in a cut in the stem of another plant. Some plants can be grown from _____, that is a piece of twig or leaf taken from a plant. These are _____ identical to the plant from which the cutting was taken. _____, for example of orchids, is another way of creating genetically identical plants.

Plants

Choose the correct response from the pair of words provided each time.

All living things are made up of (usually) small objects called (1) *cells/protoplasm*. Plant cells are different to animal cells. They have a rigid cell wall, shown by (2) A/B in the diagram, which is made of (3) *oil/cellulose*. The cell walls help support plants and hold the cells together. The cell membrane is a (4) *thick/thin* skin which lets substances in and out of the cell and is shown by (5) B/C. The (6) *nucleus/bacteria* controls the cell and is shown by (7) D/E. In the nucleus there are (8) *chromosomes/colours* which hold all the information needed to build the cell. Green plants contain a substance called chlorophyll which is contained in the (9) *cytoplasm/chloroplasts*. Chlorophyll absorbs energy from (10) *sunlight/soil* in a process called photosynthesis. Chloroplasts are shown in the diagram by (11) C/D. In the centre of a plant cell there is a large droplet of liquid — mainly (12) *air/water*. This space is called a vacuole and is shown by (13) D/E. Animal cells (14) *do/don't* have cell walls, chloroplasts or vacuoles.



Flowering plants (angiosperms) produce pollen. These small grains contain (15) *male/female* sex cells. Pollen is made in the (16) *stamens/sepals*. When pollen is carried to the female part of a flower this is called (17) *propagation /pollination*. If the pollen falls in the (18) *same/different* plant this is called self-pollination. If it happens with (19) *same/different* plants this is cross-pollination. Cross-pollination can be carried out in a number of ways. Insects, such as bees, are often attracted to flowers by their (20) *size/smell* and bright colours. In crawling through the plant to drink or collect nectar, they pick up pollen on their coats. When they visit the next flower this pollen can be transferred to a (21) *nectary/stigma*. Some birds have long (22) *beaks/claws* to reach nectar in flowers. Pollen picked up accidentally by the bird can be transferred to another plant.

Plants which rely on the wind for cross-pollination produce (23) *large/small* quantities of pollen. These plants usually have (24) *bright/dull* flowers because they don't need to attract insects. Pollen grains from wind-pollinated plants tend to be smooth and (25) *light/heavy*. Pollen from insect-pollinated plants is sticky and (26) *smooth/spiky*.

Flowering plants (27) *reproduce/change* by forming seeds. Seeds can remain dormant, that is (28) *active/inactive* for long periods of time. A seed will grow into a (29) *plant/bulb* under certain circumstances. A seed will start to grow, or (30) *dominate/germinate*, when it is at a suitable temperature, and if oxygen and (31) *water/soil* are present. Seeds store enough (32) *food/water* to help them grow. When a seed is planted, or falls, into soil its roots always grow (33) *upward/downward* and its shoot always grows (34) *upward/downward*. The roots are influenced by (35) *light/gravity* and the shoot is affected by (36) *light/gravity*. Growing towards light is called (37) *phototropism/geotropism* and growing downwards is called (38) *phototropism/geotropism*.

Some plants, (39) *annuals/biennials*, only live for one growing season. They can only reproduce (40) *once/twice* and then die when they have no more stored food. Some plants make and store food, in their roots for example. This food supply lets them live over winter to reproduce the next year and then die.

Biennials are plants which normally need (41) *one/two* years for their life cycles. Biennials store (42) *food/seeds* in the first season of their growth. They (43) *flower/mutate* in the second season and then die.

K1 WATER: ASSIGNMENT

- Very dry.
- Slightly salty.
- Water-bearing layer of permeable (porous) rock, sand, or gravel.
- A large body of ice moving across land.
- To do with water.
- Has pores or openings that allow liquids or gases to pass through.
- Relating to, or with water.
- Water, sodium chloride, magnesium chloride, calcium chloride and various bromides, iodides and sulfates, etc.
- Ozone purifies it.
 - Chlorine purifies it.
 - Fluoride strengthens teeth.
 - Alum settles floating particles.
- For example ACT: Ginini Flats, NSW: Macquarie Marshes, Gwydir Wetland, Myall Lakes, NT: Coburg Peninsula, Qld: Moreton Bay, Currawinya Lakes, Great Sandy Strait, SA: Bool and Hacks Lagoons, 'Riverland', Tas: Apsley Marshes, Jocks Lagoon, Vic: Barmah Forest, Kerang Wetlands, Gunbower Forest, WA: Ord River Flood Plain, Lake Toolibin.
- A large amount of water, specialised organisms.
- Waterfowl, frogs, mangroves, etc.
- Better described as cyanobacteria, they are microorganisms which thrive under certain conditions in water, and are poisonous.

K2 WATER SUPPLIES

- Toilet uses 70 L, personal washing uses 50 L, clothes washing uses 25 L, dishwashing uses 15 L, gardening uses 12 L and car washing uses 12 L, cooking uses 5 L, and drinking uses 3 L.
- 5 000 000 L.
- Sewage.
- Less.
- $375/1000 \times 10 \text{ L} = 3.75 \text{ L}$.
- It is pumped into large tanks, mixed with air and decomposed by bacteria.
- $50\,000 \text{ L} \times 50\,000 = 2\,500\,000\,000 \text{ L}$.
- Warmth may help fish to grow.
- Some fish, etc., may die of suffocation.
- Our average daily use is 180 L, so 10 L is about 5.5% of our consumption.
- The symptoms of cholera are diarrhoea, vomiting, thirst and muscle cramps, with death within hours of symptoms appearing. Over 50% of those infected die if not treated, less than 1% die with treatment. It is caused by bacterium *Vibrio cholerae*. Commonly spread by contaminated stools in water system. Prevention: vaccine, sanitation and clean water. Treatment: replace fluids and salts, antibiotics.

K3 AND 4 WATER

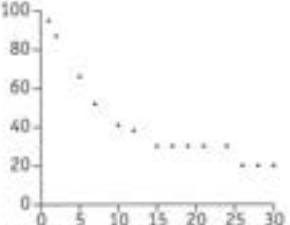
- Petrol, ethanol ('metho'), diesel, turpentine, white spirit, mercury, etc.
- Water.
- Sugar.
- Sodium chloride, NaCl.
- 35 L.
- 2.5 L.
- 140 L.
- The part of the Earth that is capable of supporting life.
- 'Saline' water contains dissolved salt, 'fresh' contains minimal dissolved substances.
- Hydrogen (H) and oxygen (O).
- 67%.
- 1 mL of water has a mass of 1 g.
- Water containing waste, etc.
- No, it is a compound containing two different elements (hydrogen and oxygen) per molecule.
- Teacher to check.
- Cirrus, cumulus, stratus and nimbus.
- Water, in rivers, etc. dissolves salt (and other minerals) on its way to the sea. Water evaporates from seas and leaves dissolved material behind. The process is repeated.
- The point is an obsolete agricultural measurement, 100 points = one inch (25 mm).
- Slightly salty water.
- Perspiration (sweating) is one way humans lose heat. In humid conditions it is more difficult for sweat to evaporate.

K5 AND 6 WATER AND LIFE

- $94/100 \times 1 \text{ kg} = 0.94 \text{ kg}$ (940 g).
- H_2O .
- O.
- CO_2 .
- The place where an organism lives.
- To do with (for instance living in) water.
- Swimming, fishing, skiing, sailing, etc.
- $78 \times 2/3 = 52 \text{ kg}$.
- $6 \text{ CO}_2 + 6 \text{ H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6 \text{ O}_2$.
- The ability to dissolve substances.
- Heat capacity.
- Expansion of ice, running water, glaciers.

- Component, part.
- Ice acts as an insulator for the water beneath.
- Heat is lost from the body when water (sweat) evaporates from the skin.
- Water can store large quantities of heat which is gained and lost at a slower rate.
- False.
- True.
- False.
- Figures vary, kangaroos use less than cattle or sheep (of equivalent mass). Hence kangaroos will survive better in dry areas and will cause less of a strain on water supplies.
- The Archimedes principle says that a submerged object has a buoyancy force (upthrust) equal to the weight of the water displaced by that object. Hence a whale will have some of its weight 'balanced' by the mass of water its body displaces.
- Seaweed, water plantain, reeds, watercress, rice, etc.

K7 WATER: SOME FACTS AND FIGURES

- $9375 = 9500 \times 100 = 98.7\%$
 - $9\,375\,000\,000 \div 350\,000 = 26\,790$ households (approximate).
 - $(250 \times 15/100) + (100 \times 50/100) = \$37.50 + \$50.00 = \87.50 .
 - $(250 \times 15/100) + (200 \times 50/100) = \$37.50 + \$100.00 = \137.50 .
 - Turn off tap..., wash car with a bucket..., only use dishwashers..., use a timer..., replace washers..., use mulch...
 - Wash car on grass..., use low-phosphorus..., collect and bury..., don't pour chemicals...
 - $45 \text{ L} \times 3 \times 52 = 7020 \text{ L}$.
 - Cost of machine, ease of use, energy efficiency, spin speed, time to wash a load etc.
- 
- $(95 - 28) = 67 \text{ g}$.
 - That the loss in mass is due to water loss, and the mass loss has finished.
 - $(67 + 95) \times 100 = 71\%$
 - Repeat the experiment, take more readings (for instance daily), use a more precise balance, extend length of time the experiment was performed.

K8 PROPERTIES OF WATER

- The solute is sodium chloride, the solvent is water.
- The processes (chemical reactions) by which a particular organism uses food for energy and growth.
- To do with (for instance living in) water.
- To do with (for instance living on) land.
- Containing salt.
- It can wear them away (rain/rivers), or freeze and expand in cracks. 'Acid rain' reacts with rocks.
- Washing, toilets, drinking, watering gardens.
- Cooling, as part of goods produced (such as lemonade), washing, solvent.
- A cubic metre (m^3) is $100 \text{ cm} \times 100 \text{ cm} \times 100 \text{ cm} = 1\,000\,000 \text{ cm}^3 = 1\,000\,000 \text{ mL}$.
Mass = $1\,000\,000 \text{ g} = 1\,000 \text{ kg}$, so density = $1\,000 \text{ kg m}^{-3}$.
- The water would freeze, and expand and could burst radiator.
- They are less dense than liquid water.
- STP = 0°C (273 K) and 101.3 kPa (1 atmosphere or 760 mm Hg).
SLC = 25°C (298 K) and 101.3 kPa (1 atmosphere or 760 mm Hg).
- It measures the amount of oxygen (in, for instance, mg L⁻¹) used by organisms and chemical reactions. BOD should be as low as possible.
- Polar (slight +/− charge) nature of dissolved substances means there is an electrostatic attraction between water molecules and dissolved particles. (See 'hydrogen bonds'.) It is harder to separate water molecules, so more energy (heat) is needed.

K9 FORCES BETWEEN WATER MOLECULES

- Viscosity.
- Cohesion.
- Cohesion.
- Surface tension.
- Adhesion (capillary action).
- Adhesion.
- Can be stretched and then resumes its former shape.
- A substance that can easily alter its shape, and takes the shape of its container. Fluids can flow. Fluids are either liquids or gases.
- When water freezes, molecules (being polar) line up to maximise attraction between positive and negative ends of the molecules. This arrangement takes up more space than the liquid form.

K10 DIFFUSION AND OSMOSIS

- 1 Water.
- 2 When a crystal dissolves in water the particles break apart, moving randomly until evenly distributed in solution. Bromine particles do the same in air.
- 3 $\text{Br}_2(\text{l})$.
- 4 To do with movement.
- 5 Oxygen moves in, carbon dioxide moves out.
- 6 When the concentrations inside and outside the cell are similar.
- 7 Right-hand side.
- 8 Water molecules will move from the left-hand side to the right-hand side.
- 9 It is semi-permeable (allows smaller particles to move through).
- 10 Carbon (C), hydrogen (H) and oxygen (O).
- 11 Similarities: both involve movement, both involve particles and the 'evening-out' of concentrations.
- 12 Differences: diffusion has solute particles moving, from high to lower concentration; osmosis is movement of solvent particles, from low to higher concentration and a membrane is involved.

K11 SUBSTANCES DISSOLVING IN WATER

- 1 $(50 \div 100) \times 1000 = 500 \text{ g L}^{-1}$
- 2 $(5 \div 500) \times 1000 = 10 \text{ g L}^{-1}$
- 3 $(25 \div 250) \times 1000 = 100 \text{ g L}^{-1}$
- 4 $(75 \div 2000) \times 1000 = 37.5 \text{ g L}^{-1}$
- 5 By stirring, using smaller particles, or using more solvent.
- 6 Something that shouldn't be there.
- 7 Cadmium (Cd), mercury (Hg), lead (Pb).
- 8 Poisonous.
- 9 Increases with subsequent additions, or 'adds up'.
- 10 The organisms of an ecological community arranged in order of predation, showing how each uses the next lower member as a food source.
- 11 Capable of being broken down in nature by microorganisms, such as bacteria.
- 12 A substance that alters rate of a reaction without itself being used up.
- 13 Usually solubility of gases in water decreases when temperature increases.

L1 KOALAS AND OTHER AUSTRALIANS

- 1 Inhabiting trees.
- 2 Listless, sluggish, passive, etc.
- 3 $(4 + 1 + 3 + 5 + 2) \div 5 = 15 \div 5 = 3 \text{ gums km}^{-2}$.
- 4 $100 \times 3 = 300 \text{ km}^2$.
- 5 Marsupial: doesn't develop a true placenta, the female has an abdominal, teat-containing pouch to carry offspring. Extinction: dying out, no longer existing. Originated: came from. Carrion: dead (rotten) flesh. Introduced: brought in from abroad. Forage: search for food. Carnivorous: meat-eating. Habitat: place where it lives. Gestation: the time for which young are carried in the uterus (womb). Offspring: children, progeny. Nocturnal: active at night.
- 6 The placenta is not well developed. The marsupial young are born less developed than most placental mammals (eutherians) very soon after conception. After birth, the young marsupial climbs into the mother's abdominal pouch (if present), or attaches itself to a teat, which expands to hold the young in place. The more developed placental mammals attach to teats only to feed.
- 7 Quolls have pointed snouts, they are of the genus *Dasyurus*, the eastern quoll has no spots on its tail, etc.
- 8 The thylacine was wolf-like or dog-like, tan in colour, with black stripes across its back.
- 9 No predators, food and shelter provided, medical treatment provided, etc.

L2 KANGAROOS

- 1 Over 50.
- 2 The giant or great grey kangaroo.
- 3 Over 6 m.
- 4 Balance and support.
- 5 Over 2.5 m.
- 6 Plants.
- 7 The grasslands and forests of eastern and southern Australia.
- 8 A gland which secretes milk.
- 9 Wallabies are smaller, and brighter in colour (and can be nocturnal).
- 10 Nocturnal: active at night.
- 11 Terrestrial: living on land.
- 12 Arboreal: living in trees.
- 13 Abundant: common, numerous.
- 14 Could include bandicoots, dunnarts, numbats, Tasmanian devils, and wombats.

L3 NOXIOUS WEEDS

- 1 Moisture, air (oxygen) and suitable temperature. (Plus the seed must be viable).
- 2 One that kills only certain plants.
- 3 The amount of time needed for a herbicide, for example, to lose its toxicity and the pasture is safe to use.
- 4 Horses, pigs (and cattle).
- 5 They might eat other (useful) plants, might displace other (useful) insects, etc.
- 6 Consult those with the problem, research the insect's natural life-cycle/predators, set up control experiments (ensuring no organisms

escape) with insects and weeds, use a variety of habitats to ensure the insect has no unforeseen effects on the Australian environment, etc.

- 7 Contaminated seed stocks, in run-off water, carried on animal fur and in animal droppings.
- 8 It is poisonous to many (stock) animals, it reduces soil fertility, and displaces wanted plants.

L4 CONCERN OVER CRAYFISH

- 1 Teacher to check.
- 2 Octopuses.
- 3 Stop catching sharks.
- 4 Too much commercial fishing, change in food supplies, change in environment, disease, etc.
- 5 Research the records of the catches of crayfish and sharks in that area, and compare them with other similar areas. Check that current fishing of crayfish is sustainable, that is that fishermen aren't catching more than can be replaced. Check current fishing methods for sharks/crayfish. Catch limited numbers of sharks and octopuses, and check their stomach contents. If the claims are true, restrict the catch of sharks.
- 6 To sell at market, to restrict numbers of sharks (because they eat other fish), to use as bait, or by accident.
- 7 Decomposers — they return decaying organic matter to food the cycle.
- 8 One not deliberately fished for, not meant to sell at market.
- 9 Tiny plant and animal life found in water.
- 10 An animal with a segmented body (arthropod) and no backbone, possessing an exoskeleton (external skeleton/shell).

L5 SURVIVAL

- 1 Food.
- 2 They will stay the same or increase.
- 3 Decrease.
- 4 Plain fish more competitive, they are faster and will beat striped fish for food.
- 5 Striped fish might survive longer, but loss of green plants might lower oxygen level making it more difficult for all fish to survive.
- 6 Same age/size/mix of sexes.
- 7 Larger fish, water fowl, humans.
- 8 Possibly insects, smaller fish, plant life.
- 9 Could include Murray Cod (*Maccullochella peelii*), Macquarie Perch (*Macquaria australasica*), Australian Smelt (*Retroinna semoni*), Barramundi (*Lates niloticus*), etc.
- 10 Billabong: a backwater that forms a pool/streambed that is usually dry but filled seasonally. Creek: stream of water, smaller than a river.

M1 AND 2 THE HEART

- 1 $70 \times 60 \times 24 \times 365 \times 70 = 2\,575\,440\,000$ beats.
- 2 Your body needs to get more oxygen and food to its cells and remove more carbon dioxide, etc.
- 3 So they can push blood further around the body.
- 4 Artificial.
- 5 Valves.
- 6 Food and oxygen (and removal of wastes).
- 7 Arteries.
- 8 Veins.
- 9 Capillaries.
- 10 A complete contraction and relaxation of the heart muscle.
- 11 300 g.
- 12 Muscle.
- 13 A pacemaker.
- 14 Reduce the amount of animal fats/cholesterol eaten, take more exercise, take drugs to reduce plaque and have surgery.
- 15 Elastic, unreactive, smooth surface.
- 16 To allow substances to pass easily through the walls.
- 17 Cost, availability of donor hearts, hospital space, suitably qualified hospital staff, etc.
- 18 This is sometimes called PTCA (percutaneous transluminal coronary angioplasty). Blockages are removed by inserting a catheter into a coronary artery, for instance, and inflating a tiny balloon. This restores blood flow.
- 19 Use of immuno-suppressive drugs, use irradiation, the donor being a close relative (unlikely in case of a heart).
- 20 Nicotine causes blood vessels to constrict and the heart to beat faster. This raises blood pressure which can lead to heart attacks and strokes.
- 21 Aspirin apparently improves blood flow in the heart's arteries during a heart attack. It seems to do this by stopping platelets clogging arteries. Figures say aspirin reduces risk of death from a heart attack by 23%.

M3 AND 4 RESPIRATION

- 1 A substance providing energy and raw materials for growth and repair.
- 2 CO_2 .
- 3 H_2O .
- 4 Gets oxygen from air, uses oxygen and food to supply energy to cells, removes unwanted gases.
- 5 Muscle.
- 6 Two.
- 7 Food and oxygen.
- 8 Out and up.
- 9 Down and in.
- 10 Oxygen relights a smouldering piece of wood.

- 11 Carbon dioxide turns limewater from clear to cloudy.
- 12 Bronchi.
- 13 Tiny blood vessels.
- 14 Bubble-shaped (spherical).
- 15 Exhaled air, contains waste carbon dioxide from cells.
- 16 True.
- 17 Observes/keeps a watch over.
- 18 Heart failure, drowning, electric shock, etc.
- 19 Cardio Pulmonary Resuscitation.
- 20 It is a device that restores and maintains breathing.
- 21 Brain damage can occur if oxygen doesn't reach the brain.
- R22 Make unconscious by lack of oxygen.
- R23 Glucose solution, plasma, blood, saline solution etc.
- R24 Refer to a first-aid book, such as the St John's Ambulance Brigade handbook. For an adult, give 15 compressions followed by two breaths, with this cycle repeated four times per minute.
- R25 Five compressions followed by one breath, with the cycle repeated 12 times per minute.
- R26 Early symptoms resemble flu, followed by permanent paralysis of the limbs.
- R27 It is a large tank enclosing the person (apart from the head), in which the pressure varies causing air to move in and out of the patient's lungs.
- R28 A vaccine was developed.
- R29 Teacher to check.

M5 THE SKIN

- 1 Epidermis (outer layer) and dermis (inner layer).
- 2 Melanin.
- 3 Touch, heat, cold, pressure and pain.
- 4 37°C.
- 5 Vitamin D.
- 6 No, it can cause sunburn, dry skin and cancers.
- 7 They are dead cells (and have no nerve endings).
- 8 Sebaceous.
- 9 Fat. Stores fat and insulates.
- 10 Minute openings, allowing passage of gases/liquids, for instance.
- 11 When the body is cold, the muscles next to hair follicles contract.
- 12 Ends of hairs are composed of dead cells, only the part in the follicle is alive, so this is a figure of speech.
- 13 Tiny blood vessels.
- 14 No, every person's fingerprint is unique.
- R15 To identify them (fingerprints might be too small).
- R16 Ultra violet radiation can damage or cause mutations in cells which can reproduce out of control.
- R17 SPF: Sun Protection Factor. The higher the number, the greater the protection. SPF 15 means that 15 minutes of exposure covered in cream would be equivalent to 1 minute of unprotected exposure. SPF 15 blocks almost all ultra violet (UV) radiation.

M6 OUR SKELETON

- 1 Heart, lung, liver, kidney, etc.
- 2 Makes things slippery, reduces friction.
- 3 Doesn't let water through it, not affected by water.
- 4 Will last a long time.
- 5 Does not react.
- 6 They connect bones.
- 7 In joints.
- 8 They are too flexible, and become misshapen over time.
- 9 Fixed joints, don't move (skull). Ball-and-socket, swivel (for example, hip). Hinge, one way motion (for example, knee). Pivot, rotation (for example, elbow). Sliding, slip (wrist and ankle).
- 10 It is durable, has low friction and doesn't react with tissue.
- 11 Tendons.
- R12 The use of engineering principles in biological and medical applications.
- R13 A mixture of metals.
- R14 Silicon compounds containing oxygen, carbon and hydrogen. Usually large molecules (polymers) that are water and heat resistant.

M7 OSTEOPOROSIS

- 1 Bones are more porous and more easily fractured than normal.
- 2 The denser the bone, the healthier it is.
- 3 It takes less time, costs less, carries less risk.
- 4 Ultrasound (sound waves that can't be heard).
- 5 The machines are smaller, cheaper, more portable and easier to operate than previous methods.
- 6 Excessive exposure to X-rays can cause damage, for example, cancers.
- 7 That the bone density in the rest of the body is the same as the heel.
- 8 No, it is quoted as a 'useful screening tool, but isn't a definitive test'. The conventional test is more accurate.
- 9 Have a high calcium intake and perform weight-bearing exercise.
- R10 Older people, post-menopausal females, those with a family history of osteoporosis, etc.
- R11 Cigarette smoking, excessive alcohol, use of corticosteroids, inactive lifestyle, low calcium intake etc.
- R12 Vitamin D (sunlight or dietary supplement), medication (such as oestrogen replacement for women).

M8 LIFESTYLE CHEMISTRY: ASSIGNMENT

- 1 A mixture of particles spread through another substance which are too small for resolution with an ordinary light microscope but can't pass through a semi-permeable membrane.

- 2 A small part or piece of something.
- 3 Slippery substance used to reduce friction between moving surfaces.
- 4 Substance used to kill pest, such as insects.
- 5 A substance (usually a liquid) capable of dissolving another substance.
- 6 When tiny drops of one liquid are spread out (but don't dissolve) and are suspended in another liquid.
- 7 It stabilises an emulsion, that is stops the droplets from separating out, or keeps them suspended.
- 8 Egg yolk (in mayonnaise), casein (in milk and cream), bile (in the stomach), etc.
- 9 It is a logarithmic scale which tells us the hydrogen ion concentration. A set of numbers which tell us whether a substance is acidic (0 to 6), neutral (7) or basic (8 to 14).
- 10 Could choose toothpaste, shampoo, indigestion tablets, etc.
- 11 pH might be related to performance, for example, toothpaste is slightly alkaline to neutralise acidity etc.
- 13 Products may include surfactant (wetting agent), colour, perfume, etc.
- 14 Tablet, liquid form, cream, injection, dermal patch, nasal spray, suppository, etc.
- 15 Mention, for example, ease of administering, packaging, quick action, slow release, etc.
- 16 To make them more attractive and palatable.

M9 SOLUBILITY OF TABLETS: PRACTICAL ASSESSMENT

(Note: ensure safety assessment is carried out prior to doing this activity.)

- The tablet has dissolved if it has disappeared (mention time taken).
- The temperature, volume of water, whether it is shaken, or stirred, etc. could all vary.
- Apparatus could include: container, clock, measuring cylinder, tablets, etc.
- Whatever you do with one tablet, do the same with the rest, to ensure the test is fair.
- Normal safety precautions should be taken, plus ensure no tablets are consumed by pupils.
- Soluble medicine would work more quickly, and would be easier to take. Could use 'normal' versus 'soluble' aspirin, various brands of paracetamol, indigestion tablets, etc.

N1 ASTRONOMY

- 1 Something in the sky/visible heavens.
- 2 Time intervals are 76, 75, 75 and 77 years, or 76 years on average. So Haley's Comet came near Earth in 1835 and 1911.
- 3 Not solid but spread out.
- 4 Everything would die.
- 5 Galaxies are moving outwards and away from us, these furthest away are moving fastest.
- 6 A sun.
- 7 A satellite orbits another (larger) celestial body. Comets orbit the sun, so are satellites.
- 8 Both have increases in brightness. A supernova is much brighter than a nova and results in destruction of the star.
- 9 False.
- 10 True.
- 11 False.
- R12 Steady state: the universe has always existed, has always been expanding and hydrogen has been created continuously. Big bang: the universe came into being billions of years ago in a huge explosion from a single point of vast density.
- R13 He studied nebulae, discovered 'red-shift' of light and, hence, helped show that galaxies are moving away from ours, etc.
- R14 He lived from 1656 to 1742, was born in London, studied the theories of Sir Isaac Newton, was made British Astronomer Royal in 1721, studied comets especially the one named after him, etc.

N2 THE DEVELOPMENT OF ASTRONOMY

- 1 For example: the study of objects outside the Earth's atmosphere.
- 2 The Geocentric theory has Earth at the centre and the Heliocentric theory has the sun at the centre of the solar system.
- 3 A celestial object with a bright head, and long tail which points away from the sun.
- 4 Copernicus.
- 5 To work out when it was time to plant crops, etc.
- 6 To do with the moon.
- 7 Earth's moon had craters, Saturn had rings and Jupiter had moons.
- 8 Teacher to check.
- R9 Accurate timekeeping was essential for working out locations of ships that had been at sea for extended periods. A large cash prize was offered by the British Board of Longitude for a sea clock that could keep good time. John Harrison made a machine which met standards but was not initially paid for his work.
- R10 Latitude gives location north or south of the equator. Longitude gives location of a place east or west of a north-south line based on Greenwich, London.
- R11 Babylonians: had a year of 12 lunar months of 30 days each, with additional months when necessary. Ancient Egyptians: had a (solar) year of 365 days, divided into 12 months of 30 days each, plus 5 days added at the end. Romans: had a calendar which changed often and eventuated in the Julian calendar. The normal year was 365 days, every fourth year (leap year) was 366 days, and gave us the months and the days of the week used today. The Gregorian calendar was introduced to remedy

discrepancies that arose. (Note: Jewish, Chinese and Islamic calendars are also used throughout the world.)

N3 ASTRONOMY AND AUSTRALIA

- 1 To observe the transit of the planet Venus.
- 2 A building to observe/study astronomical events among other things.
- 3 The study of atmosphere and weather.
- 4 For clean air and to avoid city lights.
- 5 One that detects radio waves from celestial objects.
- 6 To look for a comet.
- 7 He established an observatory and appointed two astronomers.
- 8 True.
- 9 John Tebbutt.
- 10 A place showing images of celestial objects.
- 11 John Bolton, Ronald Giovanelli, Leonard Huxley, Frederick White, Richard Woolley, etc.

N4 AND 5 ASTRONOMY ASSIGNMENT

- 1 Edmund Halley (Astronomer Royal, British, etc.)
- 2 Copernicus (named Nicolaus, was a monk, etc.)
- 3 Ptolemy (named Claudius, worked in Egypt, etc.)
- 4 Isaac Newton (was a 'Sir', worked on laws of motion, etc.)
- 5 Edwin Hubble (studied nebulae, discovered 'red shift' of light, etc.)
- 6 Neptune.
- 7 Venus.
- 8 Earth.
- 9 Jupiter.
- 10 Mars.
- 11 Saturn.
- 12 Mercury.
- 13 Teacher to check.
- 14 A small piece of celestial matter that is seen when it burns up in passing through atmosphere.
- 15 An object that orbits another (larger) celestial object.
- 16 A celestial object with a bright head, and long tail which points away from the sun.
- 17 A group of stars which forms an imaginary pattern.
- 18 A very large group of stars.

N6 AND 7 THE SOLAR SYSTEM

- 1 Nine.
- 2 The sun.
- 3 The Earth.
- 4 Mercury.
- 5 Pluto.
- 6 Jupiter.
- 8 The distances would be too large.
- 9 4880 km.
- 10 12 100 km.
- 11 12 800 km.
- 12 6800 km.
- 13 142 800 km.
- 14 120 900 km.
- 15 52 200 km.
- 16 49 400 km.
- 17 2280 km (all sizes approximate).
- 18 Mars.
- 19 Neptune.
- 20 Mercury.
- 21 Venus.
- 22 Pluto.
- 23 The sun.
- 24 Mars.
- 25 Moon.
- 26 False.
- 27 The sun.
- 28 58 000 000 km.
- 29 109 000 000 km.
- 30 149 503 000 km.
- 31 228 000 000 km.
- 32 778 3000 000 km.
- 33 1 424 000 000 km.
- 34 2 870 000 000 km.
- 35 4 500 000 000 km.
- 36 5 900 000 000 km (all distances are approximate).
- 37 Distance light travels in one year (9.5×10^{12} km approximately). Astronomical distances are huge.
- 38 The asteroid belt.

N8 TELESCOPES: AN ASSIGNMENT

- 1 Light rays are (refracted) bent by glass lenses, magnifying the image, and are brought to a focus by an eyepiece.
- 2 Earth's moon had craters, Saturn had rings and Jupiter had moons.
- 3 1564 to 1642, Italy.
- 4 Light rays are reflected and magnified by a concave mirror and brought to a focus by an eyepiece.
- 5 They are shorter, and give a better image.
- 6 Away from pollution, away from city lights, thinner atmosphere, all of which improve visibility.
- 7 Radio telescopes detect radio waves, and optical telescopes detect light from celestial objects.
- 8 They can be used during daylight, and are not affected as much by atmospheric conditions.
- 9 Named from quasi-stellar, they resemble stars, very far away and have huge energy outputs.

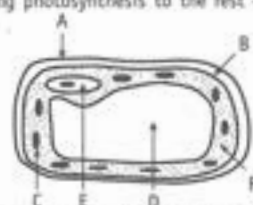
N9 ASTRONOMY: CLOZE

planets, Mercury, Pluto, solid, gas, stars, light, reflected, constellation, nebula, years, astronomers, telescopes, optical, radio, meteoroids, meteors, comets, sun, asteroids, Earth, rotation, tilted, seasons, satellite, tides, astronomical, luminosity, elements.

O1 STRUCTURE OF PLANTS

- 1 They are the reproductive organs.
- 2 The leaves.
- 3 Starch turns iodine solution from brown to blue/black.
- 4 They anchor the plant, and extract water and minerals from the soil.
- 5 Through stoma (tiny holes which open and close).
- 6 Due to transpiration — water evaporating from leaves is released by the stoma.

- 7 Plants growing towards the light.
- 8 One that makes its own food.
- 9 It uses food, and generates CO_2 .
- 10 Xylem carry water and dissolved salts from roots to leaves (transpiration). Phloem carry foods (sugars) made during photosynthesis to the rest of the plant.
- 11 False.
- 12 Diagrams will vary but should show: cell wall (A), cell membrane (B), chloroplasts (C), vacuole (D), nucleus (E) and cytoplasm (F).



- 13 A compound containing carbon, hydrogen and oxygen with hydrogen and oxygen in the ratio 2 : 1.

O2 SEEDS

- 1 Food store, embryo and seed coat (testa).
- 2 Oxygen, water and a suitable temperature.
- 3 No, they are considered to be 'resting', and will germinate if conditions are right.
- 4 To prevent them from overheating and/or germinating then dying. To keep them viable.
- 5 It protects the seed.
- 6 They may be passed their expiry date, the growing conditions may be unsuitable.
- 7 Any area which has a temperature range between 37°C and 44°C . (Plus plentiful water.)
- 8 Its tough or thick seed coat needs to be broken down before germination can occur.
- 9 The dormant seed will eventually grow, the non-viable seed will not.

O3 PLANTS: AN ASSIGNMENT

- 1 A fungus.
- 2 A conifer.
- 3 An algae.
- 4 A herbivore.
- 5 A eucalypt.
- 6 A fern.
- 7 Plant life.
- 8 Animal life.
- 9 The edible reproductive part (ripened ovary) of a seed plant, which contains seeds.
- 10 Herbaceous plant (with a soft stem) grown to be eaten.
- 11 A fruit, the edible part contains seeds.
- 12 Teacher to check.
- 13 A plant response to gravity (roots growing downwards).

O4 PLANT GROWTH AND VAN HELMONT'S EXPERIMENT

- 1 62 years.
- 2 0.1 kg.
- 3 90.9 to 90.8 that is, a drop of 0.1 kg.
- 4 The gain in the mass of the plant is due to water not the soil.
- 5 CO_2 .
- 6 Elements needed in very small amounts.
- 7 Carbon (C), hydrogen (H), oxygen (O) and nitrogen (N), sulfur (S), phosphorus (P), calcium (Ca), potassium (K), magnesium (Mg), iron (Fe), chlorine (Cl), copper (Cu), manganese (Mn), zinc (Zn), molybdenum (Mo), boron (B) and nickel (Ni).
- 8 Tomatoes, lettuce.
- 9 Plants can be grown out of season, or where they normally won't grow, etc.
- 10 Plants are supported on porous material, there are more crops per year/greater yield, etc.
- 11 Both seek knowledge and understanding of the world, philosophers tend to use speculation and thought, scientists use observation and experiment, etc.
- 11 No, metric units weren't introduced until the 1790s.
- 12 He invented the word 'gas', discovered carbon dioxide, experimented in alchemy, etc.
- 13 Teacher to check.

O5 GROWING PLANTS

dormant, viable, water, warmth, planting, leaves, food, photosynthesis, green, carbohydrates, flowers, seeds, pollination, stamens, identical, insects, self, ovary, dispersed, animals, humans, grafting, cuttings, genetically, cloning.

O6 PLANTS

- 1 Cells.
- 2 A
- 3 Cellulose.
- 4 Thin.
- 5 B
- 6 Nucleus.
- 7 E
- 8 Chromosomes.
- 9 Chloroplasts.
- 10 Sunlight.
- 11 C
- 12 Water.
- 13 D
- 14 Don't.
- 15 Male.
- 16 Stamens.
- 17 Pollination.
- 18 Same.
- 19 Different.
- 20 Smell.
- 21 Stigma.
- 22 Beaks.
- 23 Large.
- 24 Dull.
- 25 Light.
- 26 Spiky.
- 27 Reproduce.
- 28 Inactive.
- 29 Plant.
- 30 Germinate.
- 31 Water.
- 32 Food.
- 33 Downward.
- 34 Upward.
- 35 Gravity.
- 36 Light.
- 37 Phototropism.
- 38 Geotropism.
- 39 Annuals.
- 40 Once.
- 41 Two.
- 42 Food.
- 43 Flower.



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